

# ACTIVATION SPARSIFICATION IN TRANSFORMERS: PRESSURE, THRESHOLDING, AND SITE PLACEMENT

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## ABSTRACT

Activation sparsification requires choosing how to induce zeros, which operations to target, and when exploiting them improves execution. We study pressure, fixed thresholding and site placement through 54 Pythia pretraining conditions on MiniPile: extensive 14M comparisons and selected recipes at 70M and 410M, with one seed per condition. Architectural reach quantifies the multiplication workload structurally affected before training; model-wide sparsity measures observed zero-operand products using the same full-model denominator. Pressure’s marginal effect depends on threshold and targets: at 14M and  $\kappa = 0.5$ , adding OL1 to seven-site thresholding raises sparsity from 15.39% to 27.48% against 29.95% reach, for +0.1265 loss. This seven-site pressured recipe outperforms its four-site counterpart on both quality and sparsity at every tested size, while lower-threshold orderings vary. Activation distributions and operation counts explain why the broader recipe can yield more model-wide sparsity despite fewer FFN zeros. On an RTX5090, specialized BF16 full-sequence inference gains  $1.234\times$  over native execution across all 30 14M checkpoints, while sparse paths add  $1.043\times$  over matched fusion and attention skipping slows every checkpoint. These comparisons show why intervention choice should combine paired quality cost, architectural reach and execution granularity.

## 1 INTRODUCTION

Activation sparsity can reduce inference cost when specialized kernels exploit zero activations (Liu et al., 2024; Cetin et al., 2026). Choosing how to create those zeros requires knowing which operations their sites affect and whether skipping that work repays its overhead. We ask how pressure, thresholding and placement change sparsification’s quality cost and its useful computation.

Existing recipes combine these choices differently: TEAL uses post-hoc thresholding, ProSparse combines ReLU with progressive pressure, and Q-Sparse selects projection inputs (Liu et al., 2024; Song et al., 2024; Wang et al., 2024). We distinguish *pressure*, which encourages small magnitudes, *thresholding*, which applies a zeroing map, and *placement*, which selects where each acts. Fixed cutoffs let us examine their interactions while training determines how many activations survive.

We pretrain Pythia-family models from scratch for controlled comparisons across sizes (Biderman et al., 2023). At 14M, 30 conditions and 29 paired contrasts show pressure’s threshold-dependent effect: at  $\kappa = 0.5$ , adding OL1 to seven-site thresholding raises model-wide sparsity from 15.39% to 27.48% for +0.1265 loss. Architectural reach, computed before training from the graph and workload, puts this result against 29.95% structurally reachable work. A7-OL1 also improves over A4-OL1 by 14.77 sparsity pp and 0.2086 loss, despite fewer pooled FFN zeros. Selected recipes extend to 70M and 410M, where that high-threshold comparison again improves both axes, adding 5.01 and 9.02 pp; lower-threshold orderings vary. ReLU pretraining also improves the response to subsequent clipping at intermediate targets across all sizes, while retaining an initial quality cost and extreme-target reversals.

An execution case study tests which counted opportunities become useful. For BF16, batch-one, uncached 2,048-token inference on an RTX5090, the selected implementation gains  $1.234\times$  geometrically over native execution across 30 checkpoints, but sparse paths add only  $1.043\times$  over

matched fusion. Disabling attention skipping improves every checkpoint. These controls connect intervention choices to the granularity and overhead that determine latency.

## 2 RELATED WORK

### 2.1 SPARSIFICATION INTERVENTIONS

CATS thresholds small FFN activations in training-free and fine-tuning settings (Lee et al., 2024); TEAL extends magnitude thresholding to attention and FFN projections with block-wise greedy allocation (Liu et al., 2024). Our uniform clipping reference does not test TEAL’s optimized allocation. Training can also reshape the trade-off: ReLU Strikes Back restores sparse activations (Mirzadeh et al., 2023); ProSparse combines ReLU, progressive  $\ell_1$  pressure and threshold shifting, with schedule ablations showing different quality at comparable FFN sparsity (Song et al., 2024). Sparsing Law studies data, activation and architecture effects (Luo et al., 2025), and Cetin et al. (2026) induce sparse FFNs through pretraining regularization.

Beyond FFNs, Q-Sparse selects projection inputs (Wang et al., 2024), whereas Spark combines sparse FFNs with attention-position selection (You et al., 2025). Our seven-site intervention also thresholds feature coordinates inside attention. Section 3.1 explains the fixed-cutoff choice and its distinct gradient treatment. We examine pressure’s conditional effect and the resulting operation-weighted counts, with and without pressure at 14M.

### 2.2 INFERENCE SPEEDUP

TEAL and Spark report autoregressive decoding gains (Liu et al., 2024; You et al., 2025). TwELL targets batched matrix multiplication with packed sparse activations and fused FFNs, evaluating full-sequence throughput (Cetin et al., 2026). Workload and structure matter: different token masks affect batching savings; Block Q-Sparse structures masks for batched execution (Wang et al., 2024). Our case study uses execution and profiling feedback for kernel development (Ouyang et al., 2025; Luo et al., 2026), then matched fusion and skipping controls to identify what produces full-model acceleration.

## 3 METHODOLOGY

### 3.1 SPARSIFICATION INTERVENTIONS

Projection-input selection motivates train-time activation sparsification: Q-Sparse uses magnitude top- $k$  with a straight-through training rule (Wang et al., 2024). Spark addresses selection overhead with a statistical threshold estimated from each input (You et al., 2025). We test a fixed-cutoff alternative, avoiding ranking and per-input threshold estimation while leaving the surviving count free to change. Our hypothesis is that training, with optional pressure, reshapes activation mass around the cutoff. Threshold and pressure are thus separate controls; endpoint histograms examine the learned distributions.

An intervention specifies thresholding sites  $\mathcal{N}$ , their nonlinearities and thresholds, and pressure sites  $\mathcal{P}$ . The sets are independent choices, although many evaluated recipes align them (Figure 1; Appendix C.1). We use ReLU and fixed elementwise maps  $G_{+,\kappa}(x) = x\mathbf{1}\{x \geq \kappa\}$  and  $G_{\pm,\kappa}(x) = x\mathbf{1}\{|x| \geq \kappa\}$ , with  $\kappa \geq 0$ . The former removes negative values and positives below  $\kappa$ ; the latter preserves the signs and values of surviving magnitudes. At zero threshold, their forward maps are ReLU and identity. Appendix A gives boundary derivatives. We change both selection and gradient treatment relative to Q-Sparse, and do not measure a speed advantage over Spark.

Pressure uses the mean absolute activation in each targeted site-layer tensor, after its nonlinearity, then averages these means equally. Expanding the targets changes the mixture of site gradients. *Naive L1* optimizes  $\mathcal{L}_{\text{task}} + \lambda\mathcal{L}_1$ , clipping the combined gradient before AdamW.

*Orthogonal L1* (OL1) keeps task optimization primary. After task-only AdamW, it preconditions pressure using the task’s second moments, removes the component opposing the adaptive task direction under conflict, and caps the auxiliary vector relative to that direction’s norm. This adapts

gradient-conflict projection (Yu et al., 2020; Hsieh et al., 2024) while extending the norm-limiting principle of Pythia’s gradient clipping to an auxiliary update. The two caps act in different spaces.

Once this relative cap binds, increasing the pressure weight  $\lambda$  no longer increases the ideal correction. We use this conditional property to motivate a fixed pressure mechanism, holding budget  $b = 1$  and multisite  $\lambda = 1$  while varying threshold and placement. Saturation is recipe-dependent: retained logs show frequent capping for A7-OL1 at 14M and 70M, but rarely at 410M. Appendix A.2 gives the protected-direction geometry, its distinction from task-loss preservation, and the measured cap activity.

### 3.2 SPARSITY AND ARCHITECTURAL REACH

*Activation sparsity* is the exact-zero fraction at a named site. *Model-wide activation sparsity*,  $\mathcal{S}_{\text{model}} = N_0/N_{\text{model}}$ , weights zeros by affected scalar multiplications. Each declared block product gets one credit when either activation operand is zero; the denominator also includes the dense LM head, which gets no zero credit. Integer counts are pooled before division, using actual operands and causally valid pairs in full, uncached sequences (Appendix B). We report  $100\mathcal{S}_{\text{model}}\%$ ; this is multiplication opportunity, not runtime or total FLOPs.

Before training, *architectural reach* estimates how much of this workload a proposed placement can structurally affect. Formally,  $R_{\text{arch}} = N_{\text{reach}}(\mathcal{N})/N_{\text{model}}$  describes the work structurally reached if the selected sites were entirely zero, counting overlaps once. Architecture, sites and sequence length determine it, not checkpoint quality. At fixed depth and sequence length, wider Pythia blocks increase projection work quadratically in width but internal-attention work linearly; longer sequences increase the relative importance of  $QK$  and  $PV$ . Greater depth reduces the dense head’s share. These analytic consequences guide placement before spending training compute (Appendix C.2); they predict neither quality-preserving sparsity nor speedup. Natural zeros outside selected-site reach can also contribute to observed  $\mathcal{S}_{\text{model}}$ .

## 4 EXPERIMENTAL STUDY

We pretrain Pythia models from random initialization on MiniPile (Biderman et al., 2023; Kaddour, 2023): 30 conditions at 14M and 12 each at 70M and 410M. Each trains for 712 updates and 1.493 billion input tokens, with one training seed; we evaluate the final checkpoint. This common budget does not imply the same training stage at each size. Loss is next-token cross-entropy (lower is better), measured with  $\mathcal{S}_{\text{model}}$  over all 338 complete 2,048-token blocks from 500 validation documents; the 1,444-token tail is excluded. Appendix C.4 gives all settings, including the lower 410M learning rate.

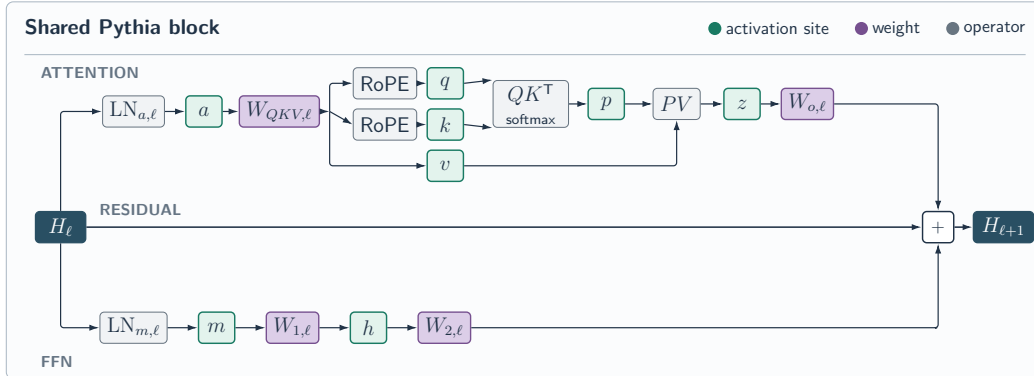


Figure 1: **Activation sites in the shared Pythia block.** A1-H replaces GELU at  $h$  with ReLU. A4 applies  $G_{+,\kappa}$  at  $a, m, h, z$ ; A7 adds  $G_{\pm,\kappa}$  at post-RoPE  $q, k$  and at  $v$ . Pressure variants target every selected site. A0 retains GELU. Appendix C.1 gives the full recipe diagram and tensor shapes.

Within each size, paired conditions share initialization, data order and budget; reused controls are not independent replications. Pressure on/off contrasts hold nonlinearities and threshold fixed. Unpressured A7 versus A4 adds symmetric  $g, k, v$  thresholding; A7-OL1 versus A4-OL1 changes complete recipes, including the target-gradient mixture and cap-dependent scale. A4 versus A1-H also changes  $h$ 's threshold above zero and its derivative at zero. Local pressure varies  $\lambda \in \{0.05, 0.1, 0.5, 1\}$ ; A4/A7 vary  $\kappa \in \{0, 0.01, 0.05, 0.1, 0.5\}$ . Multisite OL1 uses  $\lambda = 1$ ; all OL1 conditions use trust budget 1. Larger models use A0, A1-H, A4-OL1 and A7-OL1; Figure 7 lists all eight recipes.

We also apply post-hoc clipping to fixed trained checkpoints, including GELU A0 and ReLU A1-H. This second intervention uses uniform magnitude clipping at  $a, m, h, z$ , calibrated separately per site/layer on ten training blocks, for targets  $p = 0, 0.1, \dots, 0.9$ . Target  $p$  is a quantile, not achieved  $S_{\text{model}}$ . This TEAL-style control does not optimize allocation (Liu et al., 2024). It acts after the trained nonlinearity, preserving A0's GELU or A1-H's ReLU, whereas A7 replaces GELU and also thresholds internal attention. Thus the comparison changes reach and final transformations as well as intervention timing. Appendix D.6 retains all 540 evaluations.

#### 4.1 QUALITY-SPARSITY TRADE-OFFS

Local pressure and broader nonlinearities occupy different regimes (Figure 2). The lowest-loss local endpoint, A1-H-L1 at  $\lambda = 1$ , reaches 3.949% sparsity at loss 5.1023, versus A0's 5.2086. A0 clipping at  $p = 0.3$  reaches similar sparsity, 3.832%, but loss 5.3619. A7-OL1 at  $\kappa = 0.5$  instead reaches 27.483% at loss 5.8294, or +0.6208 over A0. Its 29.952% architectural reach also exceeds the four-site clipping reference's 12.833%; the comparison changes both the representation and affected computation.

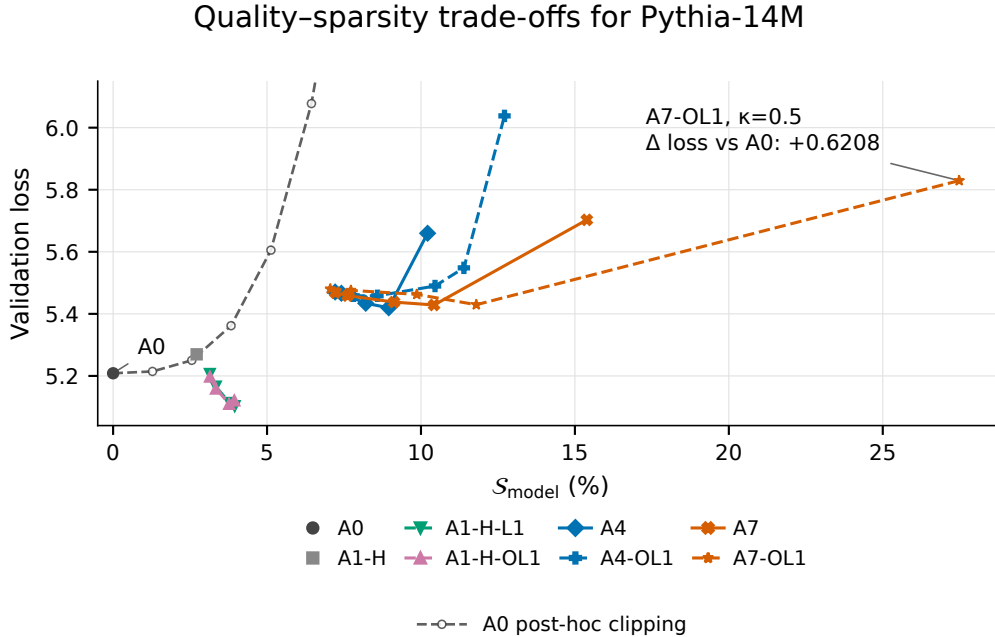


Figure 2: **Local pressure reaches lower loss; broader recipes trade quality for sparsity.** All 30 trained 14M endpoints are filled; dashed colored paths have pressure, solid paths do not. Open markers show uniform four-site A0 clipping. Six of ten clipping evaluations fall inside the displayed loss range; Appendix D.6 retains the full range. Lines join evaluated settings, not Pareto envelopes. A7 has broader reach than the clipping sites.

The learned representation also changes the response to later clipping. At 14M and matched target  $p = 0.6$ , ReLU-pretrained A1-H plus clipping reaches 7.893% sparsity at loss 6.5215, versus A0 clipping's 7.764% at 6.8212 (a 0.129 pp sparsity mismatch). Relative to each measured  $p = 0$ ,

clipping adds 1.2518 versus 1.6126 loss. This benefit overcomes ReLU’s initial +0.0611 loss cost here; it does not do so at every target. Figure 5 shows the cross-size comparison.

#### 4.2 PAIRED INTERVENTION EFFECTS

Pressure’s additional value depends on threshold and targets (Figure 3). Replacing GELU with ReLU adds 2.714 sparsity percentage points (pp) for +0.0611 loss. Local pressure lowers observed loss versus ReLU; changing naive L1 to OL1 lowers loss at three of four matched weights but reverses at  $\lambda = 1$ . This compares complete update rules, including clipping and optimizer state, rather than isolating conflict projection.

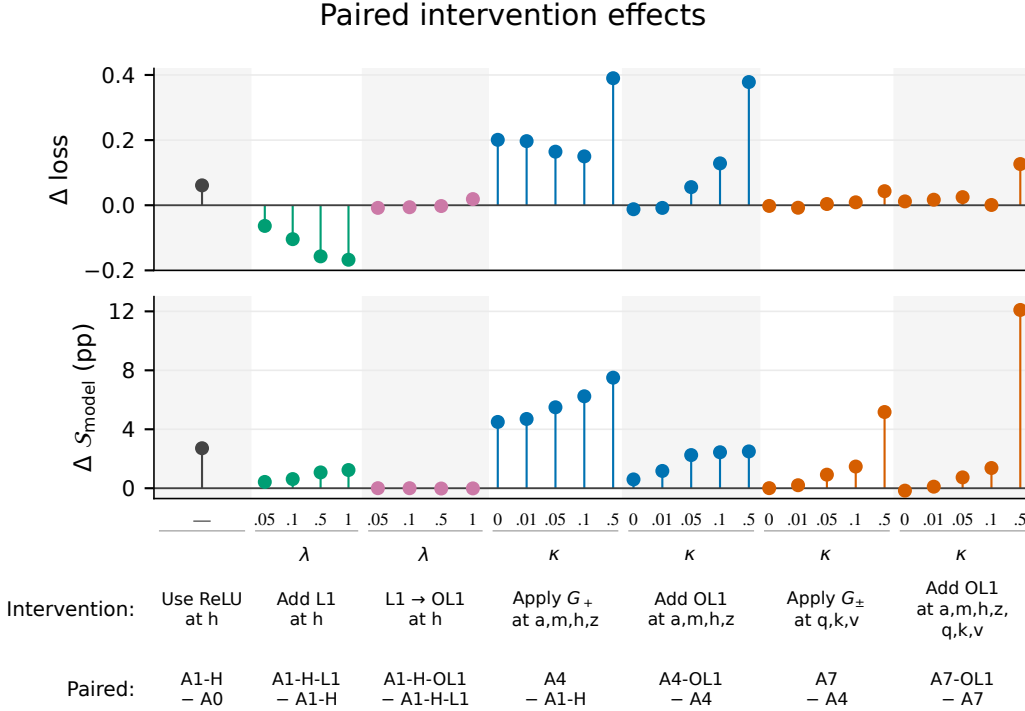


Figure 3: **Pressure’s marginal effect changes across recipes and thresholds.** All 29 paired 14M deltas are treatment minus the named reference: loss above, sparsity pp below. Labels sweep  $\lambda$  or  $\kappa$  with ordinal spacing; the dash denotes the unswept ReLU replacement. Pressure additions fix nonlinearities and threshold; other contrasts can change multiple factors (Section 4). Shared controls and one training seed do not provide 29 independent replicates. All numerical deltas are in Table 6.

For A4, adding OL1 improves both axes at  $\kappa = 0, 0.01$ , then adds sparsity at increasing quality cost: 2.498 pp for +0.3783 loss at  $\kappa = 0.5$ . The corresponding A7 addition yields 12.096 pp for +0.1265; at  $\kappa = 0.1$  it yields 1.372 pp for +0.0008, which does not establish loss equivalence. Without pressure, A7 minus A4 at  $\kappa = 0.5$  adds 5.171 pp for +0.0432 loss. The larger pressured gain under A7 changes the target-gradient mixture and, outside saturation, its scale; it cannot be assigned solely to broader thresholding. A separate historical comparison of A4 with pressure at  $h$  only (Appendix D.1) further motivates separating pressure placement from threshold placement.

#### 4.3 RESHAPING ACTIVATION DISTRIBUTIONS

More FFN zeros need not mean more model-wide sparsity (Figure 4). At  $\kappa = 0.5$ , A4-OL1 places 99.31% of pooled  $h, m$  elements at zero, versus 93.64% for A7-OL1; their attention zero fractions are 0.22% and 95.60%. A narrow nonzero density peak is therefore not a large zero probability.

Table 1: 14M operation contributions at  $\kappa = 0.5$ . All use the full-model denominator, including the dense head with no zero credit. Projections sum QKV, attention output and both FFN projections. These are counted products, not runtime or FLOPs saved.

| Recipe | Projections (pp) | $QK + PV$ (pp) | Total $\mathcal{S}_{\text{model}}$ (%) |
|--------|------------------|----------------|----------------------------------------|
| A4-OL1 | 12.657           | 0.057          | 12.713                                 |
| A7-OL1 | 10.709           | 16.773         | 27.483                                 |

A4’s untargeted attention distributions change, but comparison with A0 does not isolate pressure’s contribution; A7 intervenes there directly. Appendix D.2 retains all thresholds and tails.

### Zero mass and nonzero distributions at $\kappa = 0.5$

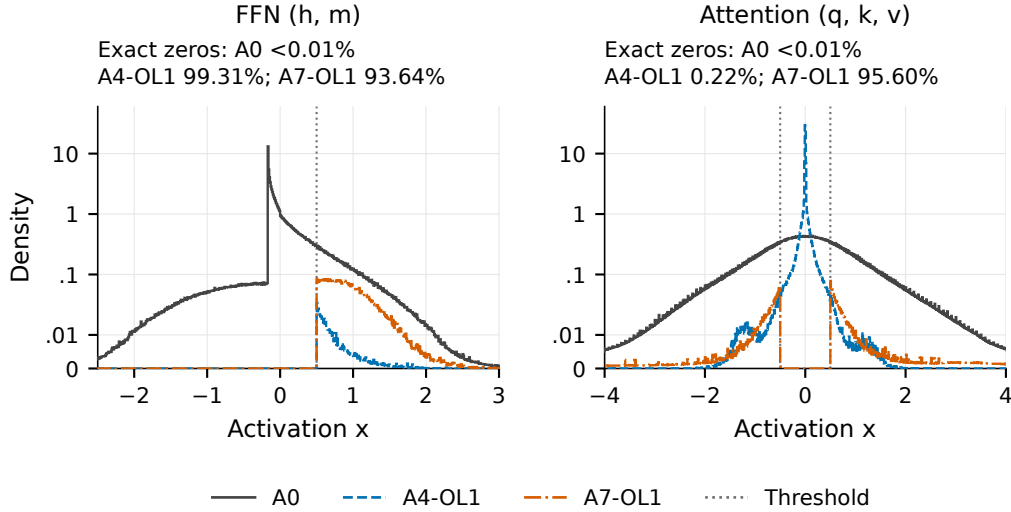


Figure 4: **A7-OL1 has fewer FFN zeros but far more attention zeros at  $\kappa = 0.5$ .** Exact-zero masses appear above nonzero densities. Counts pool all layers and 338 validation blocks, with  $h : m$  weighting 80:20 and equal  $q : k : v$  weighting, and Q/K measured after RoPE. Density divides by all elements and bin width; the symlog axis is linear below 0.01, so areas are not probabilities. Dotted lines mark thresholds. The complete grid, precision and tails are in Appendix D.2.

Operation counts explain the ranking reversal (Table 1). A7-OL1’s  $QK + PV$  contribution rises by 16.72 pp while its projection contribution falls by 1.95 pp, producing greater model-wide sparsity despite fewer FFN zeros. Reach identifies which work can be affected; the measured counts establish how much has zero operands. Local FFN sparsity alone therefore reverses the complete recipes’ apparent ranking.

In separate BF16 diagnostics, entire  $z$  rows are zero on 97.15% of rows for A4-OL1 and 91.20% for A7-OL1. On those rows the output projection returns its bias, if present. This does not zero the residual stream; whether the remaining branch contributions retain meaningful context dependence is an open question (Appendix D.3).

#### 4.4 TRANSFER ACROSS MODEL SIZES

The high-threshold recipe relationship recurs beyond 14M (Figure 5). At  $\kappa = 0.5$ , A7-OL1 improves over A4-OL1 by 0.2086, 0.1736 and 0.0703 loss and by 14.769, 5.006 and 9.024 sparsity pp at 14M, 70M and 410M. At zero threshold, A4-OL1 is better on both axes at 14M/70M, while A7-OL1 is better at 410M. Larger cohorts test complete recipes; they lack no-pressure A4/A7 controls.

A7-OL1 at  $\kappa = 0.5$  reaches 27.483%, 40.602% and 80.616% model-wide sparsity against A7 reaches of 29.952%, 49.424% and 87.245%. Part of this increase is architectural: the dense head’s

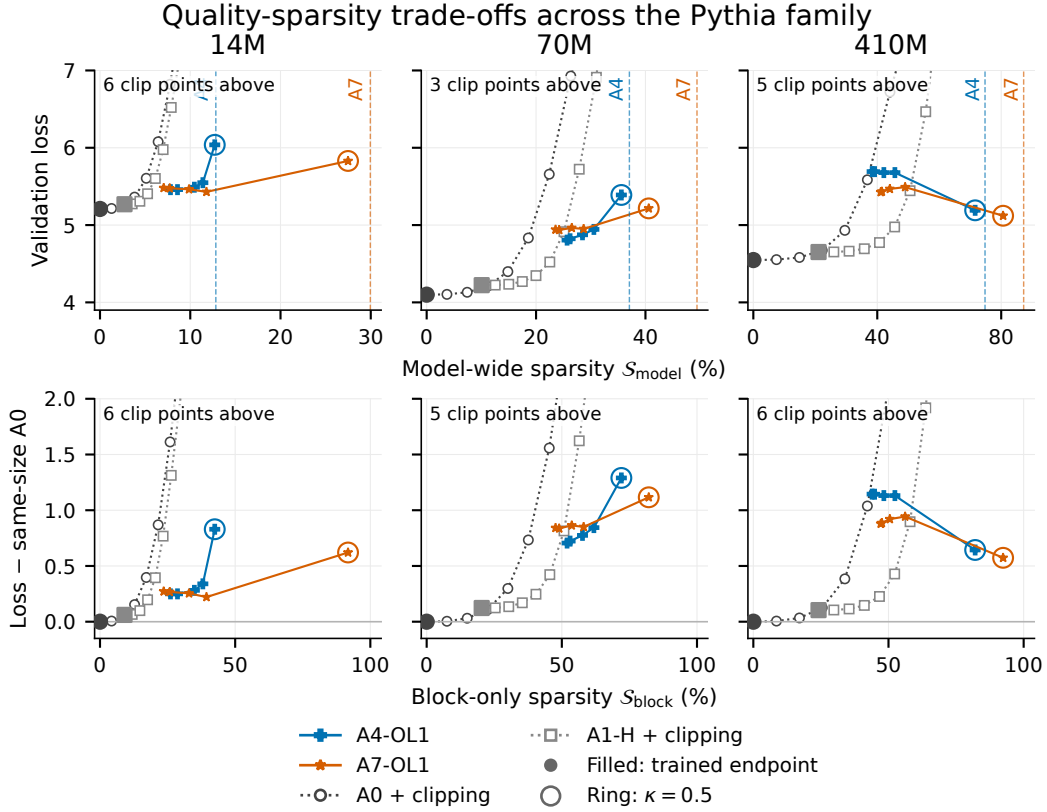


Figure 5: **High-threshold A7-OL1 improves both axes over A4-OL1 at every size.** Columns compare 14M, 70M and 410M; rows show absolute loss versus  $S_{\text{model}}$  and loss minus same-size A0 versus  $S_{\text{block}}$ . Filled colored paths connect five separately trained thresholds; rings mark  $\kappa = 0.5$ . Open paths follow ten post-hoc targets on fixed GELU A0 or ReLU A1-H checkpoints; filled gray markers show their unmodified endpoints. There are 96 distinct evaluations, repeated across rows. Dashed guides show A4/clipping and A7 reach. Top panels omit 6/3/5 high-loss clipping points; bottom panels omit 6/5/6. Appendix D.4 retains all ranges. The token budget is common, with a lower learning rate at 410M.

share falls, and the block-operation mix changes. The normalized row divides every curve by its size’s common A7 reach, giving  $S_{\text{block}}$ ; this removes head share while retaining differences in block-operation weights.

ReLU pretraining followed by clipping also has a recurring favorable region: at  $p = 0.5$ , A1-H has lower loss and higher sparsity than A0 clipping at every size. ReLU’s initial loss costs are +0.0611, +0.1230 and +0.1038, yet its incremental clipping damage is smaller at all positive tested targets. Total loss still favors A0 at  $p \leq 0.3$  and returns to favor A0 at  $p = 0.9$  for 14M and  $p \geq 0.8$  for 410M. ReLU’s starting zeros and the added four-site reach are distinct from sensitivity to removing further values; a matched quantile is not matched achieved sparsity.

Absolute quality and dense-relative penalty follow different patterns. A0 loss falls from 5.2086 to 4.0998, then rises to 4.5475 at 410M; high-threshold A7-OL1 instead improves from 5.8294 to 5.2159 to 5.1207. Its penalties over A0 are consequently +0.6208, +1.1162 and +0.5732. The 14M/70M peak learning rates agree; 410M uses a lower rate and far fewer tokens per parameter. Retained 410M A0 loss continues improving in the final 20% of training, with median pre-clipping task norm 0.850 versus 0.317/0.295 at 14M/70M (Appendix C.5). This supports investigating a finite-budget explanation, without establishing its cause.

#### 4.5 REALIZING SPARSITY WITH SPECIALIZED INFERENCE KERNELS

Which counted zeros improve full-model latency after separating fusion from skipping overhead? We compare specialized kernels with native PyTorch/SDPA CUDA graphs on one RTX5090: BF16, batch one, uncached 2,048-token inference with full vocabulary logits. Qualification checks logits and loss within specified tolerances on all 338 validation blocks. The selected K050 qualifies on all 30 checkpoints; Appendix D.7 retains the development history, TwELL compatibility and numerical contract.

Quality, latency and the conditional benefit of sparse execution

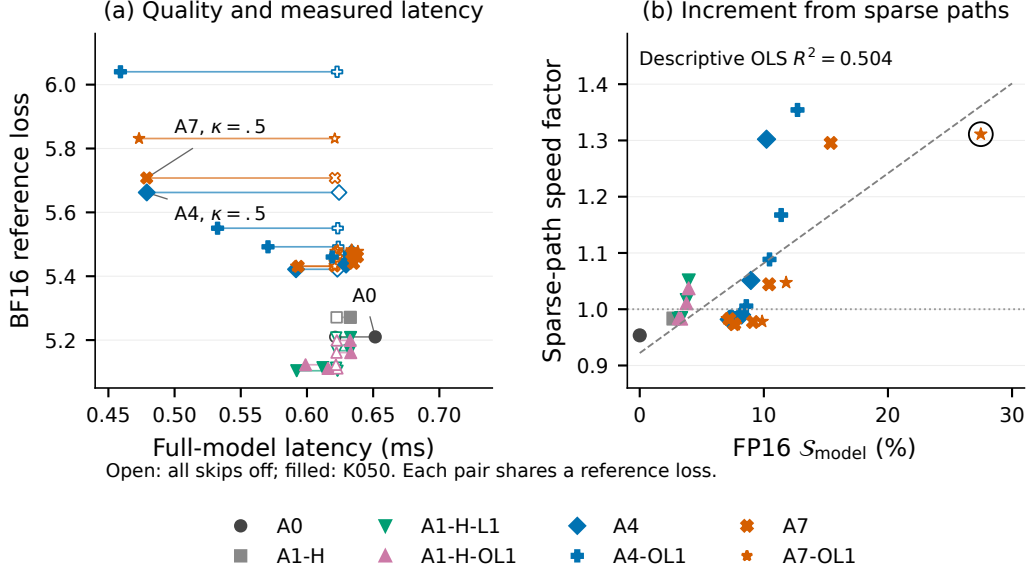


Figure 6: **More sparsity need not select a better quality–latency point.** (a) All 30 checkpoints’ native BF16 reference loss and absolute latency, paired across K050 and its all-skips-off control. Labeled A4/A7 endpoints are unpressured, at  $\kappa = 0.5$ . (b) Sparse-path factor versus canonical FP16  $S_{\text{model}}$ ; the reference line marks one and the ring marks the search checkpoint. The dashed OLS fit is descriptive. BF16 timing/quality and FP16 counts are distinct measurements; connections pair implementations on the same weights.

Speedup is native latency divided by candidate latency in paired timing samples. K050 achieves a  $1.2340\times$  equal-checkpoint geometric mean; the same implementation with all skipping disabled gives  $1.1829\times$ . The ratio of these separately measured native-relative speedups defines the sparse-path factor:  $1.0432\times$  geometrically, benefiting 14 of 30 checkpoints. At the fixed A7-OL1,  $\kappa = 0.5$  search checkpoint, the retrospective reaches  $1.7830\times$  over native and sparse paths add  $1.311\times$  over the matched control. This factor is not a direct paired latency ratio between implementations; Appendix D.7.1 retains the separate final sweep and descriptive association checks.

For example, unpressured A4 and A7 at  $\kappa = 0.5$  have specialized latencies of 0.4790 and 0.4789 ms, but BF16 losses of 5.6623 and 5.7076. A7 has more counted zeros and a larger native-relative speedup, yet the absolute latencies are almost identical and its loss is higher. Different native base-lines can enlarge a normalized speedup without identifying a better intervention on the quality–latency plane.

Disabling attention skipping improves every checkpoint, raising the native-relative geometric mean to  $1.2506\times$ . K050 bypasses an instruction when either complete operand fragment is zero; at the search checkpoint it skips 58.0% of eligible *QK* and 67.2% of *PV* instructions, yet is slower than dense attention. Partial zeros can still require the instruction, while detection and control cost time. Architectural reach justifies investigating attention; profitable execution additionally depends on fragment granularity, shape and overhead in this workload.

## 5 DISCUSSION AND CONCLUSION

Pressure’s marginal effect depends on threshold and target mixture: it can improve both quality and sparsity or exchange quality for further zeros. ReLU pretraining likewise changes the response to later clipping, with a favorable intermediate region that need not extend to extreme targets. These interactions make paired quality changes more informative than a recipe’s maximum sparsity alone.

Architectural reach helps choose placement before training by identifying the affected work. Observed counts then explain why fewer FFN zeros can coexist with greater model-wide sparsity. The high-threshold A7-OL1 advantage over A4-OL1 recurs beyond 14M, while lower-threshold orderings change. Width, depth and sequence length affect the accounting differently, so larger raw sparsity is not itself evidence of better adaptation.

Execution adds another decision: whole-fragment skips can lose despite substantial scalar zero opportunity. The evidence covers one-seed pretraining, selected larger-size recipes, uniform clipping and one 14M runtime workload. The retained trajectories motivate testing training-budget sensitivity; the cap audit motivates varying relative pressure targets rather than merely rescaling a saturated vector. Both would sharpen which intervention to train, while larger-width kernels would test when its counted zeros become profitable.

## AI USE STATEMENT

AI assistance supported code for the reported reductions, inspection of experiment records, interpretation of results, literature checks, preparation of figures and tables, and drafting and editing this manuscript. The kernel case study additionally used a human-guided coding-agent loop for implementation, testing and optimization. Its recorded default model was *gpt-6-astra*; per-request served-model identities were not verified. Reported measurements are tied to retained execution records. Numerical reductions were checked against source counts and timing samples, and kernel claims use the qualification procedure in Appendix D.7.

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## A INTERVENTION DEFINITIONS

### A.1 THRESHOLDING NONLINEARITIES

For finite  $\kappa \geq 0$ , both thresholding nonlinearities retain equality and leave retained values unchanged. Thus  $G_{+,\kappa}$  is not  $\text{ReLU}(x - \kappa)$ ; at positive cutoff both maps are discontinuous at their boundaries. The reported A4/A7 recipes use the one-sided map at  $a, m, h, z$ , including the attention projection sites  $a, z$ , and the symmetric map only at  $q, k, v$  in A7. The comparison mask is detached: the derivative with respect to the input is one for retained entries and zero for rejected entries. Consequently,  $G_{+,0}$  and  $\text{ReLU}$  have identical forward values but differ at zero, where their derivatives are one and zero, respectively.  $G_{\pm,0}$  is exactly the identity in both values and gradients. These nonlinearities use neither a straight-through estimator nor a top- $k$  selection rule. Unselected sites retain their original transformations.

### A.2 ACTIVATION PRESSURE AND OL1

For the  $J$  targeted site-layer tensors  $A_j$ , containing  $n_j$  entries each, the pressure objective is  $\mathcal{L}_1 = J^{-1} \sum_{j=1}^J n_j^{-1} \sum_{i=1}^{n_j} |A_{j,i}|$ . The tensors contain the executed site activations, after the nonlinearity where one is applied. Absolute values and means are evaluated in float32. Each tensor receives equal weight regardless of its size. Adding pressure sites changes the aggregate direction and, when the cap is inactive, its scale. Task and pressure losses are averaged over the same accumulation microbatches. For naive L1, the accumulated gradient of  $\mathcal{L}_{\text{task}} + \lambda \mathcal{L}_1$  is globally norm-clipped before AdamW (Loshchilov & Hutter, 2019).

For OL1, let  $g_t$  and  $g_1$  be the accumulated task and unweighted pressure gradients. We first clip  $g_t$  by its global norm and perform AdamW using only that clipped task gradient. Let  $\hat{m}_t$  and  $\hat{v}_t$  be its newly updated, bias-corrected moments. On parameters having both gradients and initialized optimizer state, define the adaptive task and pressure vectors  $u = \hat{m}_t / (\sqrt{\hat{v}_t} + \epsilon_A)$  and  $w = g_1 / (\sqrt{\hat{v}_t} + \epsilon_A)$ . The pressure gradient is unclipped and has no first-moment buffer. Inner products and norms below pool all these eligible parameters. With  $c = \langle u, w \rangle$ ,  $q = \|u\|_2^2$ , and numerical stabilizer  $\epsilon > 0$ , the conflict-conditioned projection is

$$\tilde{w} = \begin{cases} w - \frac{c}{q + \epsilon} u, & c < 0 \text{ and } q > \epsilon, \\ w, & \text{otherwise.} \end{cases} \quad (1)$$

For pressure weight  $\lambda \geq 0$  and required budget  $b > 0$ , define the norm ratio  $r = \lambda \|\tilde{w}\|_2 / (\|u\|_2 + \epsilon)$ . Set  $s = \min\{1, b/(r + \epsilon)\}$  when  $r > 0$ , or  $s = 1$  otherwise. After AdamW, each eligible parameter group  $p$ , with learning rate  $\alpha_p$ , receives  $\theta_p \leftarrow \theta_p - \alpha_p \lambda s \tilde{w}_p$ . The experiments' clipping thresholds,  $b$ , and  $\lambda$  are condition parameters.

The executed FP16 boundaries accumulate and unscale the two gradients separately. A nonfinite gradient skips the entire boundary, including the correction. Missing gradients or uninitialized moment state exclude a parameter from its pooled geometry. All reported OL1 runs use  $b = 1$  and  $\epsilon = 10^{-12}$ ; the local study sweeps  $\lambda \in \{0.05, 0.1, 0.5, 1\}$ , whereas multisite runs fix  $\lambda = 1$ . The norms precede group learning rates and exclude AdamW's decoupled weight decay.

**Protected direction and saturation.** For the ideal rule, omit stabilizers and take  $u \neq 0$ . Project only under conflict, using  $\tilde{w} = w - (u^\top w / \|u\|^2) u$ ; otherwise retain  $w$ . Then the pressure descent vector is

$$d_p = \min\left(\lambda, \frac{b\|u\|}{\|\tilde{w}\|}\right) \tilde{w}, \quad d_p = 0 \text{ if } \tilde{w} = 0. \quad (2)$$

It satisfies  $u^\top d_p \geq 0$ , with equality after a conflicting projection, and hence  $u^\top (u + d_p) \geq \|u\|^2$ . Also  $\|d_p\| \leq b\|u\|$ : the auxiliary vector is capped, while the combined vector can be longer than  $u$ . At a fixed gradient pair and optimizer state,  $\lambda\|\tilde{w}\| \geq b\|u\|$  makes the vector independent of further increases in  $\lambda$ . A common learning rate  $\alpha$  turns this into the added displacement  $-\alpha d_p$ .

With  $\rho = b\|u\|$  and nonzero  $\tilde{w}$ , the saturated vector maximizes  $w^\top d$  subject to  $u^\top d \geq 0$  and  $\|d\| \leq \rho$ . Without conflict this follows from Cauchy-Schwarz. Under conflict,  $w = \tilde{w} + cu/\|u\|^2$  with  $c < 0$ , so every feasible  $d$  satisfies  $w^\top d \leq \tilde{w}^\top d \leq \rho\|\tilde{w}\|$ ; the stated vector attains both

bounds. If  $w$  exactly opposes  $u$ , zero is an optimum, possibly nonunique. This is alignment with preconditioned pressure, not necessarily the first-order decrease  $g_1^\top d$ .

The protected adaptive direction generally differs from the current task gradient: first-order task loss depends on  $g_t^\top d_p$ , and finite steps also depend on curvature. Thus the property does not guarantee task-loss preservation. In the implementation, stabilizers make the projection approximately orthogonal, and  $q \leq \epsilon$  bypasses it altogether.

**Observed cap activity and objective scaling.** All 34 OL1 conditions retain all 712 boundary records, with no skipped updates. The cap is active exactly when  $r > 0$  and  $b/(r + \epsilon) < 1$ , using the logged pre-learning-rate ratio on eligible parameters. It binds on 7,283 of 24,208 boundaries (30.09%), but this pooled fraction hides large recipe differences. Across thresholds, A7-OL1 binds on 84.97–100% of steps at 14M and 100% at 70M, versus 0.42–0.84% at 410M. A4-OL1 binds on 0.14–2.67% across all sizes; local h-only OL1 ranges from 0 to 10.67% across its four weights. The first boundary has zero learning rate: its cap diagnostic describes a vector, not parameter movement. Supplementary records retain counts, correction-ratio distributions and separate early (steps 1–142) and late (570–712) summaries for every condition.

Multiplying a fixed objective by a positive scalar preserves the saturated ideal vector’s direction and norm when both versions remain saturated. For a fixed seven-site target set, replacing  $\sum_7/(7L)$  by  $\sum_7/(4L)$  only rescales that vector before the cap. Expanding four targets to seven instead changes their gradient mixture, even under saturation. Smaller coefficients in the written mean therefore do not establish proportional dilution of the effective update. Relative weights between old and new targets would test a different direction; the current experiments do not identify that contribution separately.

## B SPARSITY ACCOUNTING

### B.1 EXACT COUNTERS

Activation sparsity at site  $s$  is  $\mathcal{Z}_s = \sum_{\ell,e,i} \mathbf{1}\{A_{s,i}^{(\ell,e)} = 0\} / \sum_{\ell,e} |A_s^{(\ell,e)}|$ , where  $\ell$  indexes layers,  $e$  evaluation batches,  $i$  tensor entries, and  $|A|$  the number of elements in tensor  $A$ . Near-zero mass replaces equality by  $|A_{s,i}| \leq \epsilon$  for a stated tolerance; it does not contribute to exact-zero counts unless the value is zero. All aggregate fractions pool integer numerators and denominators across batches and layers before division.

The atomic unit of model-wide accounting is one scalar multiplication in a matrix product. If  $X \in \mathbb{R}^{n \times p}$  feeds a linear projection with  $q$  outputs, the count is  $N = npq$ , with  $N_0 = q \sum_{i,j} \mathbf{1}\{X_{ij} = 0\}$ . Zero weights receive no credit. For the multihead attention graph specified in Appendix C.1, both operands are activations, so a product is credited once if either is zero. Within one block and evaluation batch, for query position  $t$ , key position  $u \leq t$ , and within-head coordinate  $j$ ,

$$N_{0,QK} = \sum_{b,h,t} \sum_{u \leq t} \sum_j \mathbf{1}\{Q_{bhtj} = 0 \vee K_{bhuj} = 0\}, \quad (3)$$

$$N_{0,PV} = \sum_{b,h,t} \sum_{u \leq t} \sum_j \mathbf{1}\{P_{bhtu} = 0 \vee V_{bhuj} = 0\}. \quad (4)$$

Here  $P$  is the causal softmax probability tensor; the model numerator sums these counts over blocks and evaluation batches. Future masked positions are excluded from both counts, while zeros on valid positions, including floating-point underflow, are included. Counts use the operands actually consumed by each matrix product; upstream masks do not determine those operands’ zeros after intervening transformations. Earlier value positions participate in more causal pairs, so local value sparsity alone does not determine  $PV$  sparsity.

### B.2 ARCHITECTURAL REACH AND NORMALIZATION

For a specified graph and workload, let  $N_o$  count the valid scalar multiplications in block operation family  $o$ , pooled over layers and examples. Then  $N_{\text{model}} = \sum_o N_o + N_{\text{head}}$ , while  $N_0$  sums the corresponding zero-operand counts only in block operations. The block-only view is  $\mathcal{S}_{\text{block}} =$

$N_0 / \sum_o N_o$ . Both sparsities are fractions; percentages multiply them by 100. Embedding lookup, normalization, nonlinearities, softmax, RoPE, bias and residual additions, and data movement are outside this accounting.

To construct the reach numerator, count products guaranteed to have a zero activation operand as a structural consequence of 100% activation sparsity at the selected sites, independently of checkpoint values or natural zeros elsewhere. Include exact zero-preserving propagation under the declared graph rules. Products reached through more than one site receive one credit. The resulting count is  $N_{\text{reach}}(\mathcal{N})$ . Division by the same  $N_{\text{model}}$  gives  $R_{\text{arch}}$ . Natural zeros outside this selected-site reach can contribute to observed  $S_{\text{model}}$ , so architectural reach is not an upper bound on every observation. Its gap to an observation does not measure sparsity available without quality loss. Appendix C.2 specializes the denominator and reach rules to the Pythia graph used in the experiments.

## C EXPERIMENTAL SETUP DETAILS

### C.1 PYTHIA SITES AND NONLINEARITIES

Pythia uses parallel attention and FFN branches with separate LayerNorms. For batch size  $B$ , sequence length  $T$ , residual width  $d$ , FFN width  $d_f$ , and  $H$  heads, let  $d_h = d/H$ . Table 2 identifies the actual operands recorded after the selected nonlinearity. The same site mapping is applied in every layer, with one common threshold per condition. Figure 1 shows the shared block; Figure 7 below retains all recipe, pressure and reach assignments.

Table 2: Intervention sites and the matrix products that consume them directly.

| Site   | Shape                            | Consuming operation | Placement                                      |
|--------|----------------------------------|---------------------|------------------------------------------------|
| $a$    | $B \times T \times d$            | QKV projection      | After attention LayerNorm                      |
| $m$    | $B \times T \times d$            | FFN up projection   | After FFN LayerNorm                            |
| $h$    | $B \times T \times d_f$          | FFN down projection | Nonlinearity output                            |
| $q, k$ | $B \times H \times T \times d_h$ | Attention scores    | After partial RoPE                             |
| $v$    | $B \times H \times T \times d_h$ | Probability-value   | After the QKV split                            |
| $z$    | $B \times T \times d$            | Attention output    | Concatenated context, before output projection |

At  $h$ , the selected nonlinearity replaces GELU. Other sites retain their original activations unless a thresholding nonlinearity is applied. At zero threshold, A7’s additional symmetric query/key/value nonlinearities are exact identities in both values and gradients. They therefore do not change A4’s mathematical forward map in the absence of pressure. A7-OL1 still differs from A4-OL1 in its pressure targets and their equal-tensor normalization.

Query and key counts use actual post-RoPE operands: rotation can repopulate an individual zero coordinate. Likewise, the output projection uses actual post-nonlinearity context  $z$ , rather than inferring its zeros from the value mask.

### C.2 ARCHITECTURE-SPECIFIC COUNTS AND ARCHITECTURAL REACH

The operation inventory comprises QKV projection, attention scores ( $QK$ ), probability-value products ( $PV$ ), attention-output projection, and the two FFN projections. For one uncached, unpadded sequence, each block contributes  $3Td^2$  products from QKV projection,  $Td^2$  from the attention-output projection,  $Tdd_f$  from each FFN projection, and  $dT(T+1)/2$  from each of  $QK$  and  $PV$ . Thus, for  $L$  blocks and vocabulary size  $V_{\text{vocab}}$ ,

$$\begin{aligned} C_{\text{block}} &= T(4d^2 + 2dd_f) + dT(T+1), \\ C_{\text{model}} &= LC_{\text{block}} + TdV_{\text{vocab}}. \end{aligned} \quad (5)$$

Over  $B$  sequences,  $N_{\text{model}} = BC_{\text{model}}$ ; the dense head contributes only to the denominator, as defined in Section 3.2.

Let  $\mathcal{O}(\mathcal{N})$  be the union of operations reachable when the selected sites have 100% activation sparsity:  $a$  reaches QKV;  $m$  reaches FFN up;  $h$  reaches FFN down; either  $q$  or  $k$  reaches  $QK$ ;  $v$  reaches

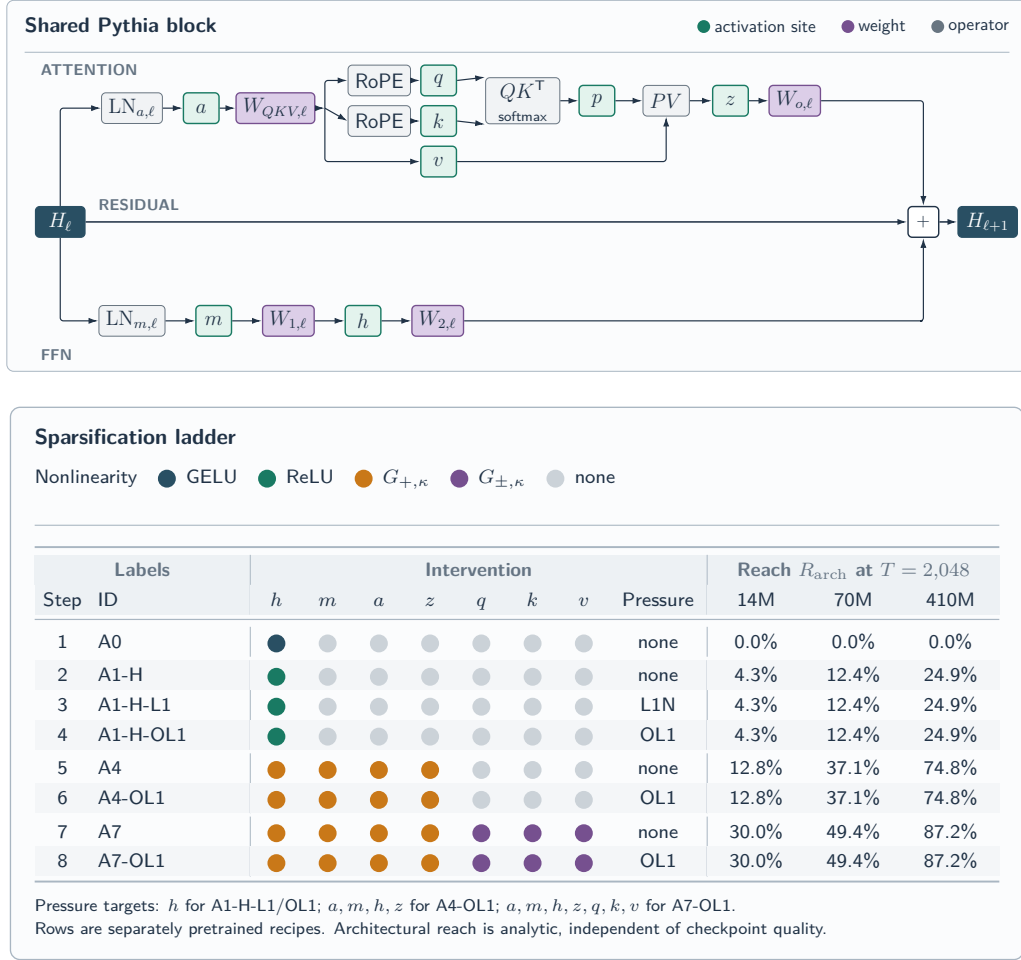


Figure 7: **Pythia architecture and sparsification recipes.** Top: the shared block and activation sites;  $q, k$  follow RoPE and  $z$  precedes the attention-output projection. Bottom: nonlinearity placement, pressure and architectural reach for the eight recipes. Selected nonlinearities replace GELU at  $h$  and act on existing activations elsewhere. L1N denotes naive L1; pressure targets every site with a selected nonlinearity in its row. The reach columns report  $100R_{\text{arch}}\%$  for  $T = 2,048$ , rather than measured sparsity or speedup. All recipes are evaluated at 14M; 70M and 410M use A0, A1-H, A4-OL1 and A7-OL1. Rows identify separately trained recipes, not curriculum stages.

PV and attention output; and  $z$  reaches attention output. Each operation is counted once. The  $v$  rule includes the exact closure  $V = 0 \Rightarrow PV = 0$ . No further propagation is assumed: QKV biases can survive  $a = 0$ , and zero scores do not imply zero softmax probabilities. For the same site set in every layer,  $N_{\text{reach}} = BL \sum_{o \in \mathcal{O}(\mathcal{N})} C_o$ , where  $C_o$  is that operation’s per-sequence count. Division by  $N_{\text{model}}$  gives  $R_{\text{arch}}$ .

With Pythia’s  $d_f = 4d$ , the per-block reachable counts for A0, A1-H, A4, and A7 are 0,  $4Td^2$ ,  $12Td^2$ , and  $C_{\text{block}}$ , respectively. The ladder evaluates these formulas at  $T = 2,048$ ,  $V_{\text{vocab}} = 50,304$ , and the pinned  $(L, d)$  configurations (6, 128), (6, 512), and (24, 1,024).<sup>1</sup> No checkpoint enters this calculation. For example, A0 has zero selected-site reach but may have positive observed sparsity from natural zeros.

Writing  $D = L(12d + T + 1) + V_{\text{vocab}}$  gives  $C_{\text{model}} = TdD$  and the planning formulas

$$R_{\text{arch}}(\{h\}) = \frac{4Ld}{D}, \quad R_{\text{arch}}(\{m, h\}) = \frac{8Ld}{D}, \quad (6)$$

$$R_{\text{arch}}(\text{A4}) = \frac{12Ld}{D}, \quad R_{\text{arch}}(\text{A7}) = \frac{L(12d + T + 1)}{D}. \quad (7)$$

The extra coverage of A7 is  $L(T + 1)/D$ , precisely the share of  $QK + PV$ ; A4 already reaches the attention-output projection. The h-only, A4 and A7 reaches are respectively 4.278%, 12.833% and 29.952% at 14M; 12.354%, 37.063% and 49.424% at 70M; and 24.925%, 74.776% and 87.245% at 410M.

At fixed  $L, T, V_{\text{vocab}}$ , projection work (including attention projections) grows as  $d^2$ , while  $QK + PV$  grows as  $d$ . At fixed architecture, projections grow as  $T$  and causal attention products as  $T(T + 1)$ , increasing internal attention’s relative importance for long sequences. Increasing depth reduces the head fraction; increasing vocabulary enlarges it. Model size therefore combines several geometric changes. These formulas describe full uncached sequences, not cached decoding. Since A7 reaches all counted block operations,  $S_{\text{block}} = S_{\text{model}}/R_{\text{arch}}(\text{A7})$ ; every curve within a size uses this same denominator.

### C.3 EVALUATION SCOPE

Quality and logical counts are paired in the same eager-attention evaluation of a checkpoint, with its declared nonlinearities and evaluation precision. The complete MiniPile validation split contains 500 documents. Concatenation and packing produce 338 complete blocks of 2,048 tokens (692,224 input tokens); the final 1,444-token tail is excluded. Each block contributes 2,047 next-token targets to cross-entropy, while product accounting includes all 2,048 input positions. Loss is averaged over the equally sized blocks, and sparsity is count-pooled. This workload uses no key-value cache. Condition-specific training budgets, optimizer settings, seeds, and checkpoint selection accompany their results; they are not implied by the ladder’s list of candidate conditions.

### C.4 REALIZED TRAINING PROTOCOL

We use MiniPile and the Pythia-14M-deduped tokenizer for every size. Each document is encoded without added special tokens, followed by one end-of-sequence token. Concatenated tokens are packed into nonoverlapping 2,048-token blocks. Training uses a seeded permutation of the block indices; validation retains source order. The training split contains 1,000,000 documents and 1,491,711,416 tokens, yielding 728,374 complete blocks and an excluded 1,464-token tail. The supplementary `protocol.json` records immutable dataset and tokenizer revisions, cache hashes and implementation identities.

All conditions start from random weights drawn from the pinned Pythia architecture configurations; released pretrained weights are not used. Model initialization and data-order seeds are both 1234, with one training seed per condition. Parameter sweeps measure intervention variability rather than variation over independent training seeds. All quality comparisons reuse the same validation split, without a separate confirmation set. Within a size, the conditions share the initial parameter hash and

<sup>1</sup>Architecture configurations: EleutherAI’s Pythia-14M-deduped, Pythia-70M-deduped, and Pythia-410M-deduped. Links identify immutable configuration revisions; released weights are not used.

training schedule. The 30-condition 14M cohort contains A0, A1-H, four weights each for A1-H-L1 and A1-H-OL1, and five thresholds each for A4, A4-OL1, A7 and A7-OL1. Each larger cohort contains A0, A1-H, and the five-threshold A4-OL1/A7-OL1 sweeps. Local pressure weights are  $\{0.05, 0.1, 0.5, 1\}$ ; thresholds are  $\{0, 0.01, 0.05, 0.1, 0.5\}$ . Multisite OL1 uses  $\lambda = 1$ , with pressure at all four or seven selected sites. Every OL1 condition uses trust budget 1 and projection stabilizer  $\epsilon = 10^{-12}$ .

Training uses AdamW with  $\beta = (0.9, 0.95)$ ,  $\epsilon = 10^{-8}$ , weight decay 0.1 and global task-gradient clipping at norm 1. Naive L1 clips the combined objective gradient instead. The learning rate warms up linearly for eight updates and follows cosine decay to one tenth of its peak. Parameters and optimizer state are FP32; forward computation uses FP16 autocast with dynamic loss scaling. Attention and hidden dropout are zero. Table 3 gives size-dependent settings. The global batch is 1,024 sequences of 2,048 tokens, for 712 optimizer updates and 1,493,172,224 input tokens. This wraps 714 blocks beyond the 728,374 complete blocks in one packed training pass. We use the final checkpoint without validation-based checkpoint selection. The lower 410M learning rate is part of the realized protocol, so size alone is not the only cross-scale change.

Table 3: Size-dependent training settings. Tokens per parameter use the common realized input-token budget; they are not held fixed across sizes.

| Setting                | 14M        | 70M        | 410M               |
|------------------------|------------|------------|--------------------|
| Parameters             | 14,067,712 | 70,426,624 | 405,334,016        |
| Peak learning rate     | $10^{-3}$  | $10^{-3}$  | $3 \times 10^{-4}$ |
| Minimum learning rate  | $10^{-4}$  | $10^{-4}$  | $3 \times 10^{-5}$ |
| Microbatch (sequences) | 32         | 4          | 4                  |
| Gradient accumulation  | 32         | 256        | 256                |
| Tokens per parameter   | 106.14     | 21.20      | 3.68               |

### C.5 RETAINED TRAINING DYNAMICS AND THE FINITE-BUDGET QUESTION

We audit A0 and A4-OL1/A7-OL1 at  $\kappa = 0.5$  at all three sizes. Each selected condition retains 712 optimizer-boundary records with no missing interval, gradient overflow or skipped update. The pre-clipping norm is the global L2 norm of the accumulated, AMP-unscaled task gradient over all trainable parameters having gradients; it precedes norm-1 clipping and AdamW. It is neither the post-clipping norm nor OL1’s eligible-subset adaptive direction norm. Task loss is the mean over a boundary’s equally sized microbatches, evaluated before its update. Figure 8 retains raw curves, aligned by actual input tokens, with the learning-rate schedules. Successful optimizer calls include the first, zero-learning-rate boundary, which produces no parameter displacement.

We fix the final 20% window, steps 570–712, before inspecting its outcomes. Table 4 gives descriptive OLS slopes of raw task loss against tokens and median pre-clipping norms over its 143 boundaries. For A0, the means of the first and last 20 records in this window change from 5.2750 to 5.2374 at 14M, 4.0506 to 4.0003 at 70M, and 4.5846 to 4.4738 at 410M. Thus all three still improve; the 410M decrease is larger. Its gradient exceeds the clipping bound on 33 of these 143 boundaries, versus none for the smaller A0s. Both 410M sparse high-threshold recipes also continue improving, whereas their 14M curves are nearly flat or slightly increasing in this window.

These observations are consistent with investigating a finite-budget limitation, but do not attribute the quality reversal to premature stopping. Raw norms also depend on parameterization, stochastic gradients and the optimizer; tokens per parameter and learning rates differ here. Only validation events at steps 1 and 712 are retained, so there is no late validation trajectory. A matched continuation of dense and sparse recipes would be needed to test whether additional training changes their ordering or their relative penalties. No such continuation is included.

Table 4: Late-run diagnostics on steps 570–712. Slopes use raw training task loss per billion input tokens, not validation loss or independent seed replicates. Sparse recipes use  $\kappa = 0.5$ .

| Size | Recipe | Median pre-clip norm | Loss slope / billion tokens |
|------|--------|----------------------|-----------------------------|
| 14M  | A0     | 0.3172               | −0.1415                     |
| 70M  | A0     | 0.2950               | −0.2194                     |
| 410M | A0     | 0.8501               | −0.4448                     |
| 14M  | A4-OL1 | 0.3651               | +0.0096                     |
| 70M  | A4-OL1 | 0.4120               | −0.0091                     |
| 410M | A4-OL1 | 0.4185               | −0.1318                     |
| 14M  | A7-OL1 | 0.3303               | +0.0207                     |
| 70M  | A7-OL1 | 0.3635               | −0.0664                     |
| 410M | A7-OL1 | 0.4429               | −0.1051                     |

Retained training trajectories (raw optimizer-boundary records)

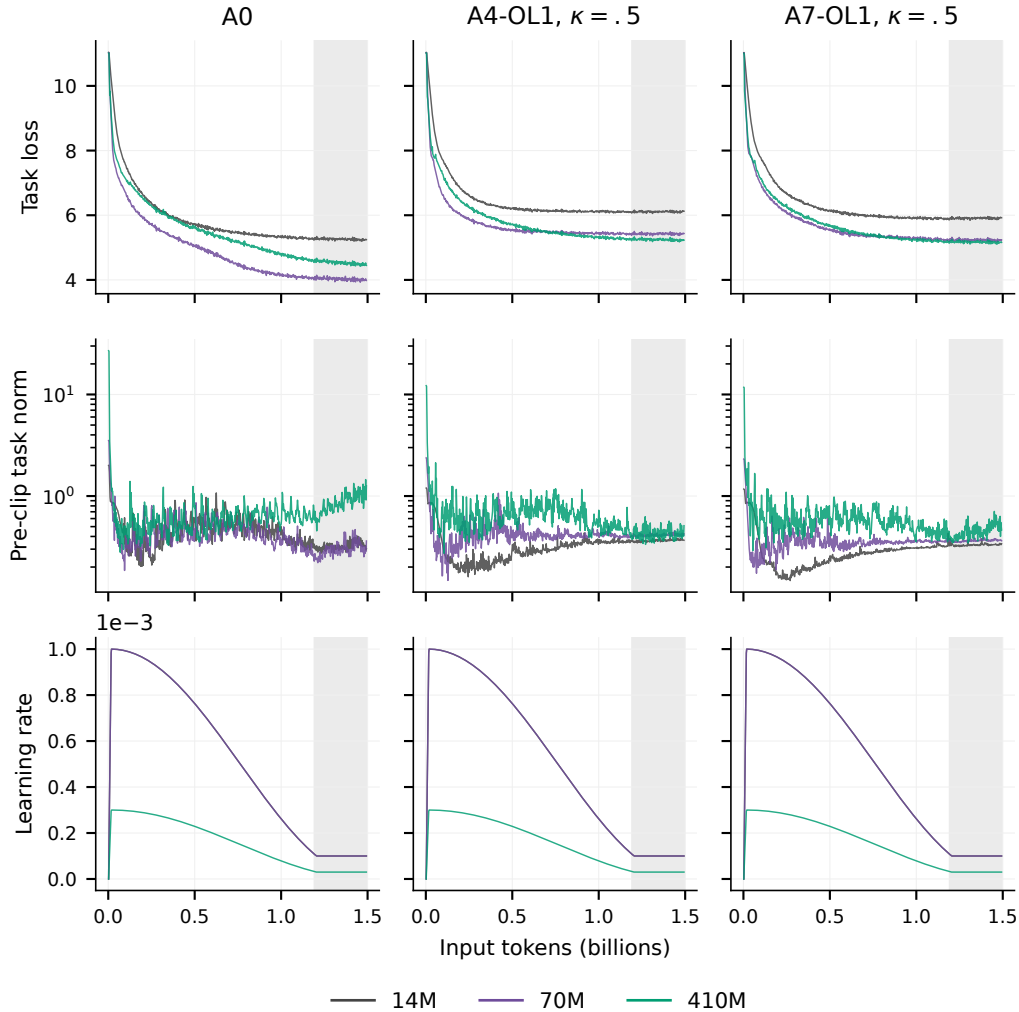


Figure 8: **410M dense and high-threshold sparse recipes continue improving late.** Columns use the same three recipes at each size; rows retain raw task loss, pre-clipping task-gradient norm (log scale), and learning rate against input tokens. No smoothing is applied. Shading identifies the fixed final-20% window. The 14M and 70M learning-rate curves coincide; 410M uses 30% of their peak and minimum rates. All nine curves contain 712 records.

## D COMPLETE RESULTS AND SUPPORTING DIAGNOSTICS

The following tables report every selected trained endpoint, all 29 paired 14M contrasts and all 15 cross-scale recipe comparisons. They retain dominated settings as well as favorable ones. Each entry summarizes one trained model on complete validation. Sparsity percentages are formed from pooled integer product counts. The endpoint table uses the common A7 architectural-reach denominator from Figure 5, rather than a different denominator for each recipe. This common normalization equals  $\mathcal{S}_{\text{block}}$ .

### D.1 A4 WITH PRESSURE AT H ONLY: A SEPARATE HISTORICAL COMPARISON

Five Run 012 checkpoints threshold  $a, m, h, z$  but apply OL1 only at  $h$ . They are outside the 54 primary conditions and the 30-checkpoint timing cohort. Their configs, inherited capture code, initialization, schedule, checkpoint hashes and complete FP16 diagnostic coverage were reconciled against Run 015’s four-site-pressure counterparts. Across the five matched thresholds, four-site pressure raises loss by 0.2425–0.3196 relative to h-only pressure. Its sparsity change is  $-0.599$  pp at zero threshold,  $+0.001$  pp at 0.01, and  $+1.421$ – $+2.486$  pp at larger thresholds. The comparison motivates choosing pressure targets separately from threshold sites: broader pressure is not automatically preferable. It changes the aggregate gradient mixture; the coefficient change from  $1/6$  to  $1/24$  alone does not identify the effective-update mechanism, because OL1’s cap can suppress global scaling (Appendix A.2). These diagnostic losses are paired with FP16 counts; terminal training losses are not substituted.

### D.2 ACTIVATION SPARSITY AND SIGNED-DENSITY COVERAGE

The seven frozen-checkpoint evaluations underlying Figure 9 record signed values after the trained nonlinearities, with Q/K measured after RoPE. Each checkpoint contributes 2,658,140,160 FFN elements and 1,594,884,096 attention elements across the complete validation set. Summing integer counts gives the 80:20 weighting of  $h : m$  and equal weighting of  $q : k : v$ . These pooled distributions are element-weighted mixtures, not distributions of layer means.

Exact zeros are stored separately from the nonzero histogram. The native grid spans  $[-8, 8]$  with 16,000 bins of nominal width 0.001; the figure combines ten bins at a time and divides by the actual floating-point bin width and the count of all captured elements. Consequently, a curve’s full integral is the recorded nonzero mass inside that grid, not one. Table 8 reports the zero mass and the fractions outside the displayed and native ranges. Underflow and overflow counts are retained rather than folded into edge bins. The displayed limits are  $[-2.5, 3]$  for FFN and  $[-4, 4]$  for attention. The largest omitted display tail is 6.6719%; the native-grid tail is at most 0.7607%.

The diagnostic reruns use FP32 parameters with FP16 autocast and no additional clipping. Their maximum absolute validation-loss difference from the retained reference is  $6.89 \times 10^{-5}$ , below the specified  $5 \times 10^{-4}$  tolerance; all 108 threshold-consistency checks pass. Small rounding differences in exact-zero counts between evaluation passes do not change the displayed conclusions. Complete per-site and pooled histograms, counts and checkpoint hashes accompany the data release.

### D.3 PROJECTION-INPUT VECTORS IN THE BF16 KERNEL DIAGNOSTIC

An untimed diagnostic pass retained the distribution of nonzero counts per token row at each site and layer for all 30 K050 checkpoints. These are BF16 kernel operands on all 338 validation blocks, separate from the FP16 histogram reruns above. Every layer contributes 692,224 rows;  $h$  has 512 coordinates per row and  $z$  has 128. Table 9 reports the two  $\kappa = 0.5$  pressured recipes. Summing the row histograms recovers the captured sitewise zero counts. Pooling rows across layers gives the all-zero-vector rates reported below. The records do not retain token identities or context perturbations, and zero projection inputs can still produce bias outputs. They therefore do not establish whether a branch has lost its function.

At  $\kappa = 0.5$ , pooling all layers gives all-zero  $h/z$  row fractions of 83.72%/97.15% for A4-OL1 and 63.12%/91.20% for A7-OL1.

Table 5: All 54 trained endpoints.  $\Delta$  loss uses same-size A0.  $\lambda$  is pressure weight;  $\kappa$  is the trained threshold. Block-only sparsity  $\mathcal{S}_{\text{block}} = \mathcal{S}_{\text{model}}/R_{\text{arch}}(\text{A7})$  uses the same A7 reference within each size, including A0. Rounded zero sparsity need not be exactly zero.

| Size | Recipe   | Parameter        | Loss   | $\Delta$ loss | $\mathcal{S}_{\text{model}}$ (%) | $\mathcal{S}_{\text{block}}$ (%) |
|------|----------|------------------|--------|---------------|----------------------------------|----------------------------------|
| 14M  | A0       | —                | 5.2086 | +0.0000       | 0.000                            | 0.000                            |
| 14M  | A1-H     | —                | 5.2696 | +0.0611       | 2.714                            | 9.061                            |
| 14M  | A1-H-L1  | $\lambda = 0.05$ | 5.2061 | -0.0024       | 3.142                            | 10.489                           |
| 14M  | A1-H-L1  | $\lambda = 0.1$  | 5.1655 | -0.0431       | 3.336                            | 11.139                           |
| 14M  | A1-H-L1  | $\lambda = 0.5$  | 5.1127 | -0.0959       | 3.788                            | 12.645                           |
| 14M  | A1-H-L1  | $\lambda = 1$    | 5.1023 | -0.1063       | 3.949                            | 13.185                           |
| 14M  | A1-H-OL1 | $\lambda = 0.05$ | 5.1981 | -0.0105       | 3.145                            | 10.501                           |
| 14M  | A1-H-OL1 | $\lambda = 0.1$  | 5.1594 | -0.0492       | 3.339                            | 11.146                           |
| 14M  | A1-H-OL1 | $\lambda = 0.5$  | 5.1102 | -0.0983       | 3.768                            | 12.580                           |
| 14M  | A1-H-OL1 | $\lambda = 1$    | 5.1212 | -0.0874       | 3.938                            | 13.149                           |
| 14M  | A4       | $\kappa = 0$     | 5.4705 | +0.2619       | 7.212                            | 24.078                           |
| 14M  | A4       | $\kappa = 0.01$  | 5.4665 | +0.2579       | 7.414                            | 24.752                           |
| 14M  | A4       | $\kappa = 0.05$  | 5.4341 | +0.2255       | 8.206                            | 27.396                           |
| 14M  | A4       | $\kappa = 0.1$   | 5.4197 | +0.2111       | 8.953                            | 29.891                           |
| 14M  | A4       | $\kappa = 0.5$   | 5.6597 | +0.4511       | 10.216                           | 34.106                           |
| 14M  | A4-OL1   | $\kappa = 0$     | 5.4583 | +0.2497       | 7.810                            | 26.075                           |
| 14M  | A4-OL1   | $\kappa = 0.01$  | 5.4583 | +0.2497       | 8.587                            | 28.668                           |
| 14M  | A4-OL1   | $\kappa = 0.05$  | 5.4898 | +0.2812       | 10.456                           | 34.910                           |
| 14M  | A4-OL1   | $\kappa = 0.1$   | 5.5483 | +0.3397       | 11.394                           | 38.041                           |
| 14M  | A4-OL1   | $\kappa = 0.5$   | 6.0380 | +0.8294       | 12.713                           | 42.446                           |
| 14M  | A7       | $\kappa = 0$     | 5.4684 | +0.2598       | 7.218                            | 24.097                           |
| 14M  | A7       | $\kappa = 0.01$  | 5.4588 | +0.2502       | 7.618                            | 25.433                           |
| 14M  | A7       | $\kappa = 0.05$  | 5.4379 | +0.2293       | 9.127                            | 30.471                           |
| 14M  | A7       | $\kappa = 0.1$   | 5.4287 | +0.2201       | 10.425                           | 34.805                           |
| 14M  | A7       | $\kappa = 0.5$   | 5.7029 | +0.4943       | 15.387                           | 51.371                           |
| 14M  | A7-OL1   | $\kappa = 0$     | 5.4802 | +0.2716       | 7.054                            | 23.551                           |
| 14M  | A7-OL1   | $\kappa = 0.01$  | 5.4758 | +0.2672       | 7.723                            | 25.786                           |
| 14M  | A7-OL1   | $\kappa = 0.05$  | 5.4628 | +0.2542       | 9.863                            | 32.929                           |
| 14M  | A7-OL1   | $\kappa = 0.1$   | 5.4295 | +0.2209       | 11.797                           | 39.385                           |
| 14M  | A7-OL1   | $\kappa = 0.5$   | 5.8294 | +0.6208       | 27.483                           | 91.755                           |
| 70M  | A0       | —                | 4.0998 | +0.0000       | 0.000                            | 0.001                            |
| 70M  | A1-H     | —                | 4.2227 | +0.1230       | 10.066                           | 20.366                           |
| 70M  | A4-OL1   | $\kappa = 0$     | 4.8054 | +0.7056       | 25.673                           | 51.944                           |
| 70M  | A4-OL1   | $\kappa = 0.01$  | 4.8229 | +0.7231       | 26.224                           | 53.060                           |
| 70M  | A4-OL1   | $\kappa = 0.05$  | 4.8742 | +0.7744       | 28.490                           | 57.644                           |
| 70M  | A4-OL1   | $\kappa = 0.1$   | 4.9439 | +0.8441       | 30.574                           | 61.861                           |
| 70M  | A4-OL1   | $\kappa = 0.5$   | 5.3895 | +1.2897       | 35.596                           | 72.022                           |
| 70M  | A7-OL1   | $\kappa = 0$     | 4.9412 | +0.8414       | 23.562                           | 47.674                           |
| 70M  | A7-OL1   | $\kappa = 0.01$  | 4.9364 | +0.8366       | 24.226                           | 49.017                           |
| 70M  | A7-OL1   | $\kappa = 0.05$  | 4.9637 | +0.8639       | 26.524                           | 53.666                           |
| 70M  | A7-OL1   | $\kappa = 0.1$   | 4.9501 | +0.8503       | 28.694                           | 58.056                           |
| 70M  | A7-OL1   | $\kappa = 0.5$   | 5.2159 | +1.1162       | 40.602                           | 82.150                           |
| 410M | A0       | —                | 4.5475 | +0.0000       | 0.011                            | 0.013                            |
| 410M | A1-H     | —                | 4.6513 | +0.1038       | 21.050                           | 24.127                           |
| 410M | A4-OL1   | $\kappa = 0$     | 5.6921 | +1.1446       | 38.339                           | 43.944                           |
| 410M | A4-OL1   | $\kappa = 0.01$  | 5.6914 | +1.1439       | 39.149                           | 44.873                           |
| 410M | A4-OL1   | $\kappa = 0.05$  | 5.6789 | +1.1315       | 42.125                           | 48.283                           |
| 410M | A4-OL1   | $\kappa = 0.1$   | 5.6789 | +1.1314       | 45.688                           | 52.367                           |
| 410M | A4-OL1   | $\kappa = 0.5$   | 5.1910 | +0.6435       | 71.591                           | 82.058                           |
| 410M | A7-OL1   | $\kappa = 0$     | 5.4263 | +0.8788       | 41.121                           | 47.133                           |
| 410M | A7-OL1   | $\kappa = 0.01$  | 5.4331 | +0.8857       | 41.478                           | 47.542                           |
| 410M | A7-OL1   | $\kappa = 0.05$  | 5.4679 | +0.9204       | 44.053                           | 50.494                           |
| 410M | A7-OL1   | $\kappa = 0.1$   | 5.4907 | +0.9433       | 48.998                           | 56.161                           |
| 410M | A7-OL1   | $\kappa = 0.5$   | 5.1207 | +0.5732       | 80.616                           | 92.401                           |

Table 6: All 29 Pythia-14M contrasts, treatment minus reference. When both recipes have a swept parameter, its value is matched; A0 and A1-H are fixed controls.

| Treatment – reference | Parameter        | $\Delta$ loss | $\Delta\mathcal{S}_{\text{model}}$ (pp) |
|-----------------------|------------------|---------------|-----------------------------------------|
| A1-H – A0             | —                | +0.0611       | +2.714                                  |
| A1-H-L1 – A1-H        | $\lambda = 0.05$ | -0.0635       | +0.427                                  |
| A1-H-L1 – A1-H        | $\lambda = 0.1$  | -0.1042       | +0.622                                  |
| A1-H-L1 – A1-H        | $\lambda = 0.5$  | -0.1569       | +1.073                                  |
| A1-H-L1 – A1-H        | $\lambda = 1$    | -0.1674       | +1.235                                  |
| A1-H-OL1 – A1-H-L1    | $\lambda = 0.05$ | -0.0081       | +0.004                                  |
| A1-H-OL1 – A1-H-L1    | $\lambda = 0.1$  | -0.0061       | +0.002                                  |
| A1-H-OL1 – A1-H-L1    | $\lambda = 0.5$  | -0.0025       | -0.020                                  |
| A1-H-OL1 – A1-H-L1    | $\lambda = 1$    | +0.0189       | -0.011                                  |
| A4 – A1-H             | $\kappa = 0$     | +0.2009       | +4.498                                  |
| A4 – A1-H             | $\kappa = 0.01$  | +0.1969       | +4.700                                  |
| A4 – A1-H             | $\kappa = 0.05$  | +0.1645       | +5.492                                  |
| A4 – A1-H             | $\kappa = 0.1$   | +0.1500       | +6.239                                  |
| A4 – A1-H             | $\kappa = 0.5$   | +0.3900       | +7.501                                  |
| A4-OL1 – A4           | $\kappa = 0$     | -0.0122       | +0.598                                  |
| A4-OL1 – A4           | $\kappa = 0.01$  | -0.0082       | +1.173                                  |
| A4-OL1 – A4           | $\kappa = 0.05$  | +0.0556       | +2.251                                  |
| A4-OL1 – A4           | $\kappa = 0.1$   | +0.1286       | +2.441                                  |
| A4-OL1 – A4           | $\kappa = 0.5$   | +0.3783       | +2.498                                  |
| A7 – A4               | $\kappa = 0$     | -0.0021       | +0.006                                  |
| A7 – A4               | $\kappa = 0.01$  | -0.0077       | +0.204                                  |
| A7 – A4               | $\kappa = 0.05$  | +0.0038       | +0.921                                  |
| A7 – A4               | $\kappa = 0.1$   | +0.0090       | +1.472                                  |
| A7 – A4               | $\kappa = 0.5$   | +0.0432       | +5.171                                  |
| A7-OL1 – A7           | $\kappa = 0$     | +0.0118       | -0.163                                  |
| A7-OL1 – A7           | $\kappa = 0.01$  | +0.0170       | +0.106                                  |
| A7-OL1 – A7           | $\kappa = 0.05$  | +0.0249       | +0.736                                  |
| A7-OL1 – A7           | $\kappa = 0.1$   | +0.0008       | +1.372                                  |
| A7-OL1 – A7           | $\kappa = 0.5$   | +0.1265       | +12.096                                 |

Table 7: All matched A7-OL1 minus A4-OL1 comparisons. Negative loss changes and positive sparsity changes favor A7-OL1.

| Size | $\kappa$ | $\Delta$ loss | $\Delta\mathcal{S}_{\text{model}}$ (pp) |
|------|----------|---------------|-----------------------------------------|
| 14M  | 0        | +0.0219       | -0.756                                  |
| 14M  | 0.01     | +0.0175       | -0.863                                  |
| 14M  | 0.05     | -0.0270       | -0.593                                  |
| 14M  | 0.1      | -0.1188       | +0.402                                  |
| 14M  | 0.5      | -0.2086       | +14.769                                 |
| 70M  | 0        | +0.1358       | -2.110                                  |
| 70M  | 0.01     | +0.1135       | -1.998                                  |
| 70M  | 0.05     | +0.0895       | -1.966                                  |
| 70M  | 0.1      | +0.0062       | -1.880                                  |
| 70M  | 0.5      | -0.1736       | +5.006                                  |
| 410M | 0        | -0.2658       | +2.783                                  |
| 410M | 0.01     | -0.2582       | +2.329                                  |
| 410M | 0.05     | -0.2111       | +1.929                                  |
| 410M | 0.1      | -0.1881       | +3.310                                  |
| 410M | 0.5      | -0.0703       | +9.024                                  |

Table 8: Activation sparsity (the probability mass at zero) omitted from the density curves, and mass outside their displayed range (view tail) and the native  $[-8, 8]$  histogram (grid tail). Grid tails are a subset of view tails, not an additional mass to add. All percentages divide integer counts by all captured elements. A0 is listed once although it is repeated in each figure row. These 14 groups come from seven checkpoints.

| Recipe | $\kappa$ | Group     | Sparsity (%) | View tail (%) | Grid tail (%) |
|--------|----------|-----------|--------------|---------------|---------------|
| A0     | —        | FFN       | < 0.01       | 0.1113        | 0.0000        |
| A0     | —        | Attention | < 0.01       | 2.3929        | 0.4632        |
| A4-OL1 | 0        | FFN       | 67.22        | 0.0003        | 0.0000        |
| A4-OL1 | 0        | Attention | < 0.01       | 5.1717        | 0.6162        |
| A4-OL1 | 0.05     | FFN       | 86.71        | 0.0003        | 0.0000        |
| A4-OL1 | 0.05     | Attention | < 0.01       | 5.6601        | 0.7607        |
| A4-OL1 | 0.5      | FFN       | 99.31        | 0.0001        | 0.0000        |
| A4-OL1 | 0.5      | Attention | 0.22         | 0.0003        | 0.0000        |
| A7-OL1 | 0        | FFN       | 64.52        | 0.0063        | 0.0000        |
| A7-OL1 | 0        | Attention | < 0.01       | 5.4657        | 0.4648        |
| A7-OL1 | 0.05     | FFN       | 75.29        | 0.0106        | 0.0000        |
| A7-OL1 | 0.05     | Attention | 7.26         | 6.6719        | 0.6932        |
| A7-OL1 | 0.5      | FFN       | 93.64        | 0.0037        | 0.0000        |
| A7-OL1 | 0.5      | Attention | 95.60        | 0.2396        | 0.0009        |

Table 9: Entirely zero projection-input rows and mean nonzero coordinates per row (NNZ), separately at  $h$  and  $z$ , for the 14M  $\kappa = 0.5$  endpoints. Layer indices start at zero. All entries use actual BF16 operands in the untimed K050 diagnostic; they are not the pooled FP16  $h, m$  histograms.

| Recipe | Layer | $h = 0$ rows (%) | Mean NNZ $h$ | $z = 0$ rows (%) | Mean NNZ $z$ |
|--------|-------|------------------|--------------|------------------|--------------|
| A4-OL1 | 0     | 12.45            | 1.729        | 90.34            | 0.116        |
| A4-OL1 | 1     | 99.63            | 0.004        | 100.00           | 0.000        |
| A4-OL1 | 2     | 100.00           | 0.000        | 100.00           | 0.000        |
| A4-OL1 | 3     | 99.91            | 0.001        | 100.00           | 0.000        |
| A4-OL1 | 4     | 99.59            | 0.011        | 92.59            | 0.074        |
| A4-OL1 | 5     | 90.77            | 0.114        | 100.00           | 0.000        |
| A7-OL1 | 0     | 22.60            | 2.022        | 69.09            | 0.405        |
| A7-OL1 | 1     | 85.01            | 0.182        | 96.38            | 0.045        |
| A7-OL1 | 2     | 75.10            | 0.296        | 95.72            | 0.047        |
| A7-OL1 | 3     | 78.99            | 0.263        | 87.77            | 0.123        |
| A7-OL1 | 4     | 62.28            | 0.526        | 98.24            | 0.018        |
| A7-OL1 | 5     | 54.77            | 0.536        | 100.00           | 0.000        |

## Exact-zero mass and nonzero activation distributions

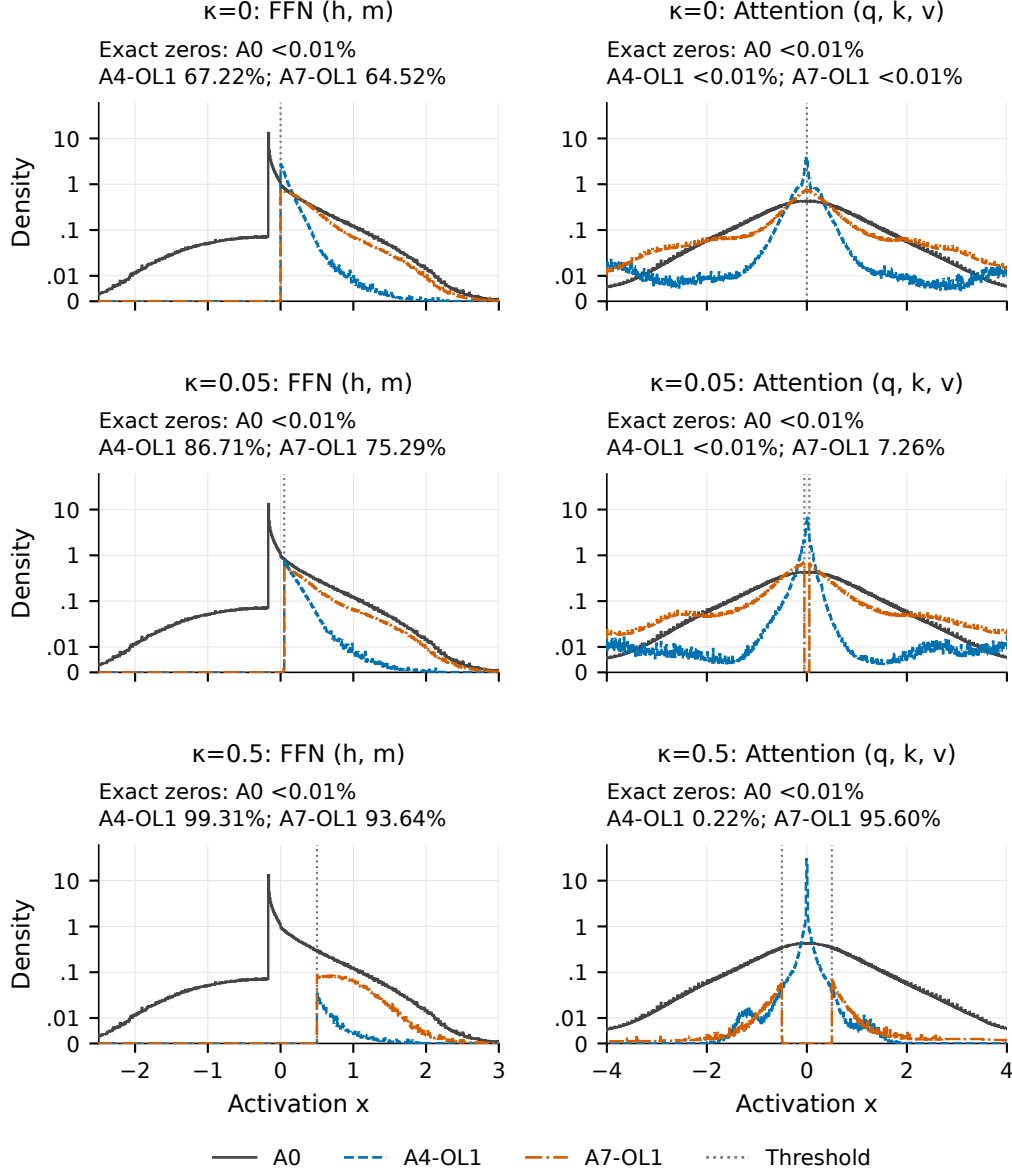


Figure 9: **Interventions reshape FFN and attention distributions differently. Exact zeros are reported above each panel and excluded from the nonzero curves.** Columns pool FFN ( $h, m$ ) and attention ( $q, k, v$ ) activations; rows use trained thresholds  $\kappa = 0, 0.05, 0.5$ . A0 is repeated across rows. Counts are pooled across all layers and validation blocks, weighting  $h : m$  by 80:20 and  $q : k : v$  equally. Density divides nonzero bin counts by all captured elements and bin width, without renormalizing the visible range. The vertical scale is symlog, linear below 0.01 and logarithmic above it; plotted areas do not represent probability mass. Dotted lines mark  $+\kappa$  for A4/A7 FFN thresholds and  $\pm\kappa$  for A7 attention thresholds. Q/K are measured after RoPE. A4 leaves  $q, k, v$  without direct thresholding or pressure; A7 targets them directly. The seven checkpoints use no additional post-hoc clipping.



Figure 10: **Complete cross-size trajectories, including extreme-target reversals.** All 96 evaluations from Figure 5 appear in both rows, without loss-range omissions. Filled colored markers connect separately trained A4-OL1/A7-OL1 thresholds; rings mark  $\kappa = 0.5$ . Open paths follow A0/A1-H through ten clipping targets, using measured  $p = 0$ ; filled gray markers denote their canonical unmodified endpoints. Every lower-row curve uses the common  $S_{\text{block}} = S_{\text{model}}/R_{\text{arch}}(\text{A7})$  normalization. Raw-view guides show A4/clipping and A7 reach; the former also applies to A1-H after four-site clipping. All 15 A7-OL1 minus A4-OL1 contrasts are in Table 7.

#### D.4 COMPLETE CROSS-SIZE CURVES AND NORMALIZATION

Figure 10 expands the main cross-size figure to its full loss ranges, including the GELU A0 and ReLU A1-H clipping tails. There are 36 trained endpoints and 60 measured clipping evaluations, repeated across both rows. The larger 396-record comparison dataset additionally retains clipping of every selected A4-OL1/A7-OL1 checkpoint; all 540 clipping evaluations across the complete primary study remain below.

#### D.5 OPERATION CONTRIBUTIONS TO MODEL-WIDE SPARSITY

An operation’s contribution to  $S_{\text{model}}$  is its zero-product count divided by the full model product count. Summing these contributions recovers the model-wide fraction; summing local activation percentages would not. Table 10 gives this decomposition at  $\kappa = 0.5$ . At 14M, extending from A4-OL1 to A7-OL1 adds attention opportunity while reducing the contribution from some projections. At larger sizes, projection work carries more weight in the workload, so an equal change in activation sparsity need not yield an equal change in model-wide sparsity. Figure 11 displays the operation-level view at the same threshold and all three sizes.

Table 10: Contributions to  $100S_{\text{model}}$  in percentage points at  $\kappa = 0.5$ . All six operations share the same model denominator within each size. The dense LM head contributes to that denominator but receives no zero credit. Differences in rounded entries can affect the displayed sum.

| Operation        | 14M    |        | 70M    |        | 410M   |        |
|------------------|--------|--------|--------|--------|--------|--------|
|                  | A4-OL1 | A7-OL1 | A4-OL1 | A7-OL1 | A4-OL1 | A7-OL1 |
| QKV              | 3.173  | 2.430  | 8.820  | 6.231  | 17.563 | 16.992 |
| $QK$             | 0.000  | 8.324  | 0.000  | 5.052  | 0.000  | 5.206  |
| $PV$             | 0.057  | 8.449  | 0.011  | 5.371  | 0.008  | 5.858  |
| Attention output | 1.069  | 1.069  | 3.087  | 3.084  | 6.186  | 6.187  |
| FFN up           | 4.139  | 2.939  | 11.329 | 8.526  | 22.938 | 21.503 |
| FFN down         | 4.275  | 4.272  | 12.350 | 12.338 | 24.895 | 24.869 |
| Total            | 12.713 | 27.483 | 35.596 | 40.602 | 71.591 | 80.616 |

Operation contributions at  $\kappa = .5$

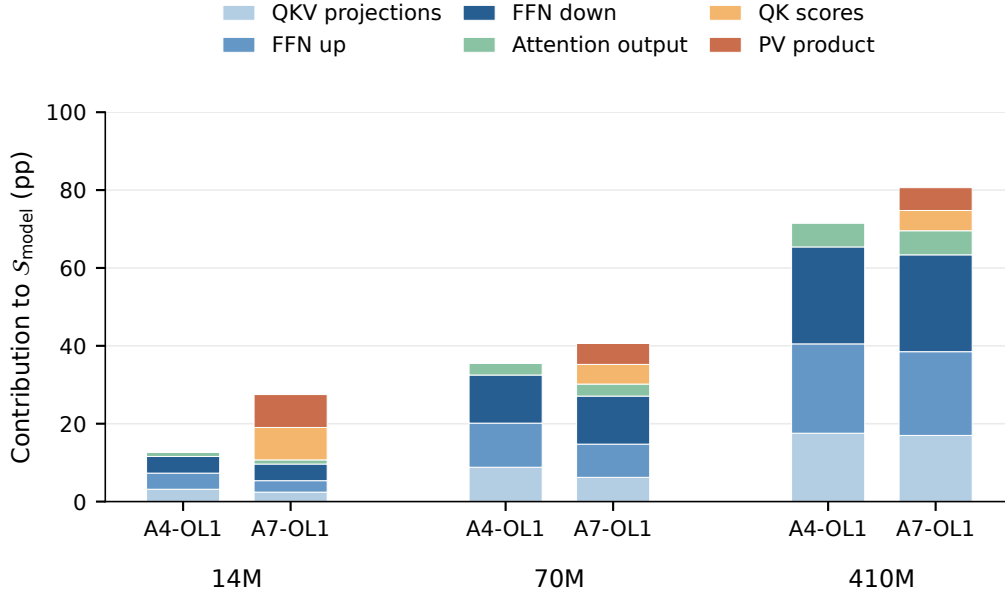


Figure 11: Operation accounting for A4-OL1 and A7-OL1 at  $\kappa = 0.5$  across 14M, 70M and 410M. Zero-operand products are counted over causally valid positions and pooled across all 338 validation blocks. Each contribution uses the full model denominator, including the dense LM head without zero credit. At 14M, attention contributes more under A7-OL1, outweighing smaller contributions from several projections. The bars show logical multiplication opportunities, not runtime savings.

## D.6 COMPLETE POST-HOC CLIPPING TRAJECTORIES

The complete release contains 540 evaluations: ten clipping targets for each of the 54 trained checkpoints. It joins 350 new evaluations with 190 retained evaluations matched by checkpoint content hash. Calibration uses the first ten complete training blocks in source order, separately for each layer’s  $a, m, h, z$  activations. Evaluation zeros values satisfying  $|x| \leq t$  after the trained nonlinearities, without changing weights or A7’s internal attention nonlinearities. The target  $p$  controls a calibration quantile; it is not the achieved model-wide sparsity. In particular, an existing zero mass can make several targets identical.

Figures 12–14 retain the full loss range and every target, including dominated points and ties. Each path begins at its actually measured  $p = 0$  evaluation. Small numerical differences from the canonical training endpoint can change strict nondominance without implying a substantive improvement. Across all checkpoints, the absolute  $p = 0$  discrepancy is at most  $1.81 \times 10^{-4}$  in loss and 0.000908 sparsity percentage points. Checking dominance beyond the observed same-size drift on both axes still finds a trained endpoint with lower loss and higher sparsity than each A0 clipping evaluation at  $p \geq 0.5$ , for every size. This is a numerical sensitivity check, not a confidence interval or an estimate of training variability. Numerical frontier memberships are stored separately from the lines connecting evaluated settings. All evaluations use complete validation with FP32 parameters, FP16 autocast, eager attention, batch one and  $T = 2,048$ .

For completeness, Table 11 retains the retrospective loss-budget reduction. It selects observed points, without interpolation; clipped selections have no matched latency measurement.

Table 11: Maximum observed sparsity under same-size A0 loss plus the stated budget, across 54 trained and 540 clipped evaluations. Trained excludes A0; “none” means no eligible sparse recipe. Budgets are retrospective; identical selections are grouped.

| Size | Loss budget      | Trained recipe        | $\mathcal{S}_{\text{model}}$ (%) | Including clipping             | $\mathcal{S}_{\text{model}}$ (%) |
|------|------------------|-----------------------|----------------------------------|--------------------------------|----------------------------------|
| 14M  | 0.05, 0.10, 0.20 | A1-H-L1 $\lambda = 1$ | 3.949                            | A1-H-L1 $\lambda = 1, p = 0.3$ | 6.490                            |
| 70M  | 0.05, 0.10       | none                  | —                                | A0, $p = 0.2$                  | 7.410                            |
| 70M  | 0.20             | A1-H                  | 10.066                           | A1-H, $p = 0.3$                | 17.485                           |
| 410M | 0.05, 0.10       | none                  | —                                | A0, $p = 0.2$                  | 14.883                           |
| 410M | 0.20             | A1-H                  | 21.050                           | A1-H, $p = 0.3$                | 35.810                           |

14M training and post-hoc clipping

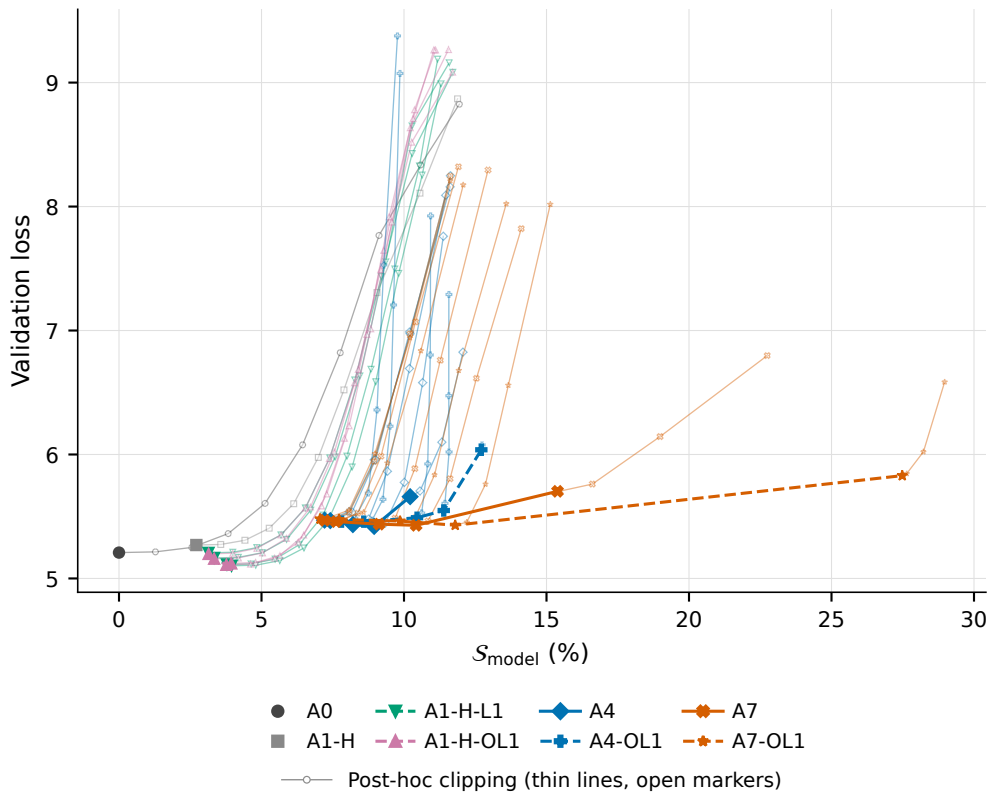


Figure 12: Complete 14M training and post-hoc clipping results: 30 trained endpoints and 300 clipping evaluations. Large filled markers denote trained endpoints; bold lines connect their settings, solid for unpressured recipes and dashed for L1/OL1. Each faint open-marker path follows one fixed checkpoint through  $p = 0, 0.1, \dots, 0.9$ . Colors and symbols identify its source recipe. The full loss range is shown. Connections order evaluated settings and do not interpolate attainable models or fitted Pareto envelopes.

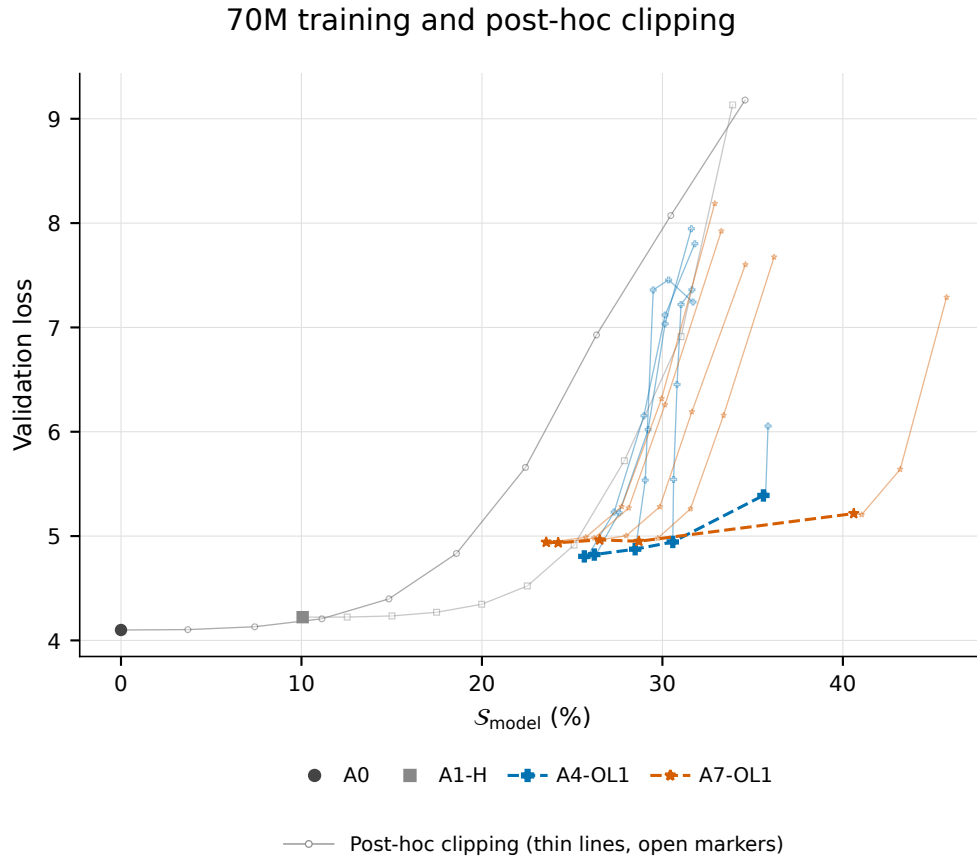


Figure 13: Complete 70M training and post-hoc clipping results: 12 trained endpoints and 120 clipping evaluations. Large filled markers and bold lines identify the trained recipes; faint open-marker paths follow each checkpoint through ten increasing clipping targets. All points, including overlaps and high-loss settings, are retained. The evaluation and line conventions match Figure 12.

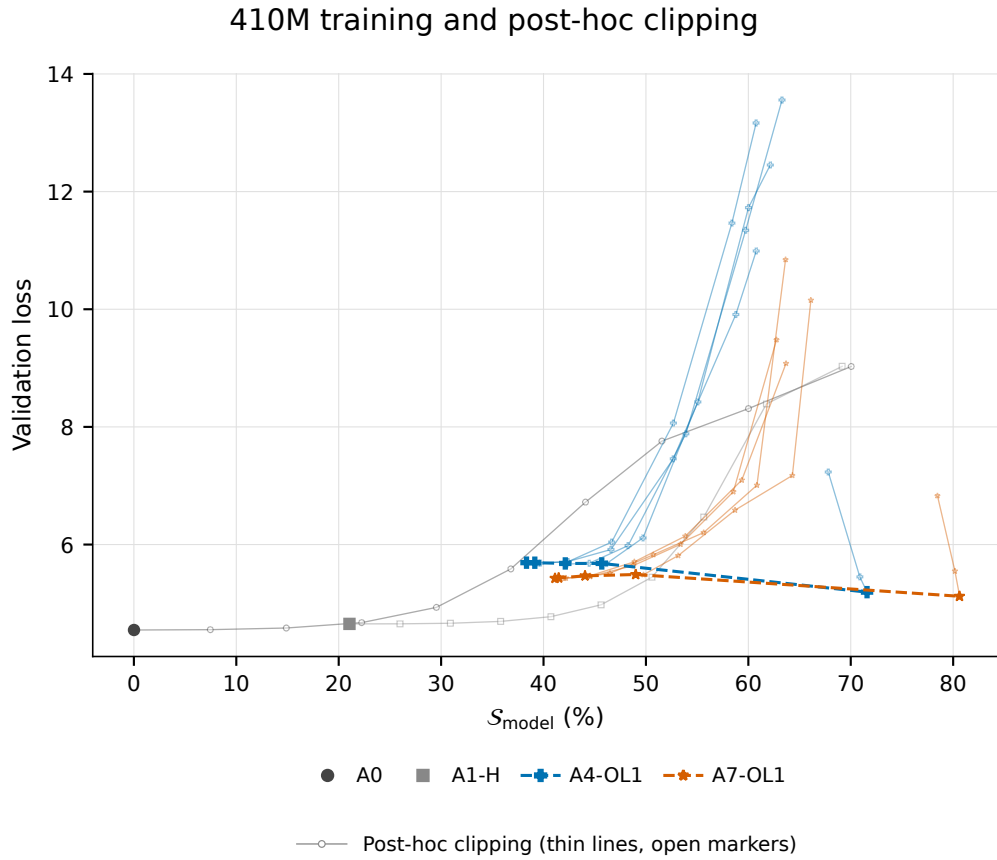


Figure 14: Complete 410M training and post-hoc clipping results: 12 trained endpoints and 120 clipping evaluations. Large filled markers and bold lines identify the trained recipes; faint open-marker paths follow each checkpoint through ten increasing clipping targets. All points, including overlaps and high-loss settings, are retained. The evaluation and line conventions match Figure 12.

## D.7 KERNEL QUALIFICATION AND ABLATIONS

The unchanged Sakana/TwELL control achieves  $1.3001\times$  on its supported SparseLM-0.5B/H100 workload. Its raw projection primitive at the Pythia-14M width  $N = 128$  instead gives relative L2 error 0.69055, exceeding the 0.02 bound; that test stops before full-model Pythia timing. This is a compatibility test, not an unchanged upstream speedup comparison on our model. The later P0 adapter changes signed-value packing, shape tails and numerical handling for Pythia. It retains dense attention and uses native projection fallback on A0. Table 12 reports this adapted baseline.

Historical proposals are replayed on four representative checkpoints; the plotted incumbent uses fixed A7-OL1,  $\kappa = 0.5$ . Final implementation and ablation summaries retain the 30-checkpoint manuscript cohort from the historical 35-checkpoint sweep. Five historical A4 checkpoints with pressure only at  $h$  remain outside that cohort; Appendix D.1 gives the separate scientific comparison. Search progress and the final sweep are separate measurements, explaining the small difference between their  $1.7830\times$  and  $1.7832\times$  maxima.

### Matched retrospective search and native-relative association

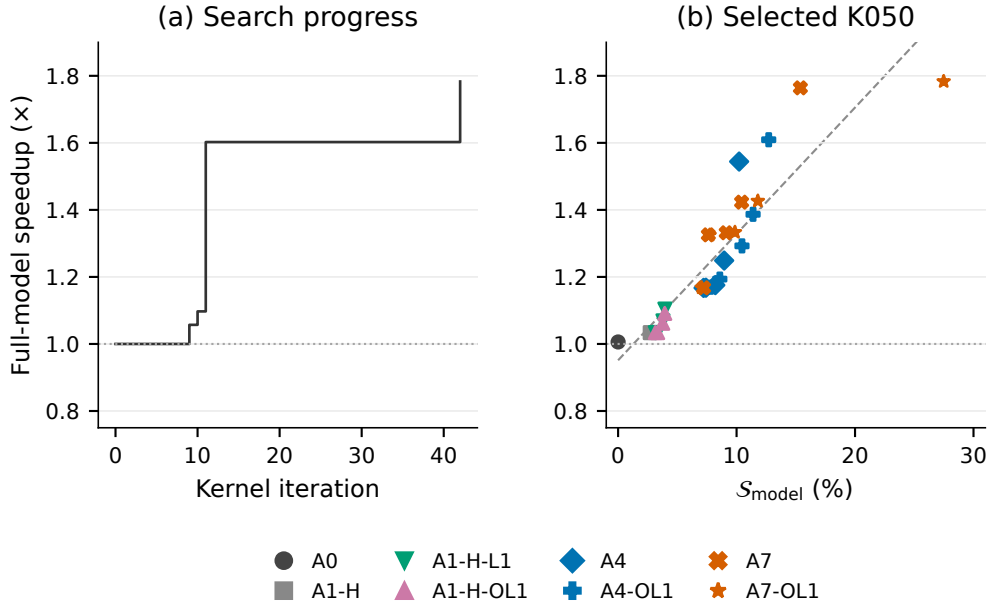


Figure 15: Retained development history and descriptive native-relative association. (a) Best qualified incumbent on the fixed A7-OL1,  $\kappa = 0.5$  search checkpoint:  $1.6024\times$  by iteration 11 and  $1.7830\times$  at iteration 42. The later attention-dense ablation is not substituted into this history. (b) Selected K050 on all 30 canonical checkpoints, with an OLS fit and intercept ( $R^2 = 0.8167$ ). Sparsity uses FP16 counts and timing uses BF16; the axes have restricted ranges. Both panels use the matched RTX5090, batch-one, full-sequence native reference specified in the text.

The coding-agent configuration records *gpt-6-astra* as the default model, not the model served for every request. This is one human-guided development trajectory and does not compare agent-assisted with human-only kernel development. Kernel measurements use Python 3.12.3, PyTorch 2.11.0+cu128, Transformers 5.12.1, CUDA 12.8 and nvcc 12.8.93. The supplementary protocol records dependency and frozen-source hashes.

Every candidate must produce finite outputs, with each logit satisfying  $|y - \hat{y}| \leq 0.25 + 0.02|\hat{y}|$ , per-block relative output L2 error at most 0.02, and absolute pooled loss difference at most 0.001. Here  $\hat{y}$  is the unmodified native eager reference. All three fresh processes must pass on all 338 validation blocks. Qualification is bounded numerical agreement on these inputs, not bitwise equivalence. Failed candidates are excluded from qualified speedup summaries. P0, an early projection adapter

derived from the released kernels of Cetin et al. (2026), qualifies on only four checkpoints. Its average is not comparable to a 30-checkpoint mean or an unchanged upstream benchmark.

Timing uses the same checkpoint, inputs and CUDA-graph scaffold for native PyTorch/SDPA and each candidate. For each checkpoint, we take the geometric mean of 1,344 paired host-latency ratios (native divided by candidate): 64 inputs, seven randomized paired passes, and three fresh processes. Compilation, static weight preparation, graph capture and equal input staging are excluded. Preprocessing that depends on activations, embeddings, every block, final normalization and full vocabulary logits are included. There is no activation cache. Cohort summaries give each checkpoint equal weight.

K050 combines normalization and RoPE fusion, zero-fragment tensor-core bypass and sparse-row projection paths, using CUDA, CUTLASS, FlashAttention and PyTorch components. K049 is its searched predecessor; K050 adds fusion of the two input LayerNorms while preserving their separate affine parameters and thresholds. The all-skips-off control retains fusion while disabling both attention and projection skips; the attention-dense control retains projection skips. Their ratios are formed from separately measured, native-normalized timings. K050’s sparse paths improve 14 of 30 checkpoints relative to all-skips-off, with geometric mean ratio  $1.0432\times$ . Disabling attention skipping improves all 30: at the search checkpoint, instrumentation confirms that skipping bypasses 58.0% of eligible  $QK$  and 67.2% of  $PV$  tensor-core matrix multiply-accumulate (MMA) instructions, yet its overhead exceeds the saved time. Scalar zero-product fractions and skipped MMA fractions are different estimands.

The attention path tests whether either warp-distributed operand fragment contains only zeros, and issues an MMA only when both fragments contain a nonzero. Partial sparsity can therefore leave an entire instruction active. Zero detection, warp votes and branches recur inside the kernel; causal masking and softmax still execute. A zero  $QK$  score does not imply a zero softmax probability. The measurements do not identify a profitable joint-zero pattern or break-even threshold. They show that the skipped work does not repay these costs for the measured 14M shapes and hardware.

Table 12: Numerically qualified full-model speedups against native SDPA CUDA graphs. GM is the equal-checkpoint geometric mean; the range is across qualified checkpoints, not a confidence interval. All conditions use one RTX5090, BF16, batch one, 2,048 tokens and full vocabulary logits. P0’s four-checkpoint subset is shown only to document its limited qualification.

| Implementation             | Qualified | GM speedup | Range         |
|----------------------------|-----------|------------|---------------|
| K049                       | 30/30     | 1.1141     | 0.9550–1.5284 |
| K050                       | 30/30     | 1.2340     | 1.0059–1.7832 |
| K050, attention dense      | 30/30     | 1.2506     | 1.0185–1.8022 |
| K050, all skips off        | 30/30     | 1.1829     | 1.0503–1.3628 |
| P0 (qualified subset only) | 4/30      | 0.8910     | 0.8180–1.0012 |

#### D.7.1 ABSOLUTE LATENCIES AND ASSOCIATION CHECKS

Table 13 retains actual host latencies: for each implementation and checkpoint, the geometric mean of the three process medians, each over 448 timed sequences. The native column uses K050’s paired native measurements; the ablation columns have their own native-paired processes. A ratio of these median summaries need not equal the reported geometric mean of paired ratios. The last column instead divides K050’s native-relative speedup by all-skips-off’s native-relative speedup. BF16 validation losses accompany these timings; canonical training tables retain their original FP16 loss/count pass.

The association checks in Table 14 fit an unweighted checkpoint-level line with an intercept. Removing the searched checkpoint or a recipe family tests descriptive sensitivity; it does not create a held-out prediction study. Within-family fits use only four or five points, and their larger  $R^2$  values do not remove the differences among recipes. All fits compare canonical FP16  $S_{\text{model}}$  with BF16 timing. The retained BF16 scalar counter excludes probability-underflow zeros and is only a lower bound, so it cannot replace the canonical counter to resolve that precision mismatch.

Table 13: Matched 14M host latency summaries in milliseconds, BF16 validation loss and incremental sparse-path factor. Dose is  $\lambda$  for local pressure and  $\kappa$  for A4/A7. All four implementations qualify on every row. These are unmodified trained checkpoints; clipped endpoints have no corresponding latency measurements. The sparse factor is defined in the text, not computed by dividing the displayed rounded latency columns.

| ID  | Recipe   | Dose | BF16 loss | Native | Skips off | K050   | Attn. dense | Factor |
|-----|----------|------|-----------|--------|-----------|--------|-------------|--------|
| c01 | A0       | —    | 5.2100    | 0.6555 | 0.6213    | 0.6516 | 0.6437      | 0.9535 |
| c02 | A1-H     | —    | 5.2712    | 0.6539 | 0.6224    | 0.6328 | 0.6242      | 0.9835 |
| c03 | A1-H-L1  | 0.05 | 5.2082    | 0.6541 | 0.6221    | 0.6329 | 0.6237      | 0.9828 |
| c04 | A1-H-L1  | 0.1  | 5.1668    | 0.6539 | 0.6226    | 0.6324 | 0.6240      | 0.9843 |
| c05 | A1-H-L1  | 0.5  | 5.1136    | 0.6541 | 0.6220    | 0.6120 | 0.6044      | 1.0162 |
| c06 | A1-H-L1  | 1    | 5.1040    | 0.6543 | 0.6231    | 0.5922 | 0.5838      | 1.0512 |
| c07 | A1-H-OL1 | 0.05 | 5.1991    | 0.6540 | 0.6225    | 0.6326 | 0.6254      | 0.9843 |
| c08 | A1-H-OL1 | 0.1  | 5.1609    | 0.6540 | 0.6223    | 0.6331 | 0.6247      | 0.9833 |
| c09 | A1-H-OL1 | 0.5  | 5.1114    | 0.6542 | 0.6230    | 0.6159 | 0.6082      | 1.0106 |
| c10 | A1-H-OL1 | 1    | 5.1228    | 0.6540 | 0.6219    | 0.5989 | 0.5909      | 1.0372 |
| c11 | A4       | 0    | 5.4727    | 0.7399 | 0.6234    | 0.6338 | 0.6250      | 0.9820 |
| c12 | A4       | 0.01 | 5.4694    | 0.7402 | 0.6234    | 0.6339 | 0.6251      | 0.9839 |
| c13 | A4       | 0.05 | 5.4364    | 0.7400 | 0.6233    | 0.6295 | 0.6213      | 0.9898 |
| c14 | A4       | 0.1  | 5.4218    | 0.7399 | 0.6230    | 0.5917 | 0.5836      | 1.0511 |
| c15 | A4       | 0.5  | 5.6623    | 0.7394 | 0.6242    | 0.4790 | 0.4713      | 1.3022 |
| c16 | A4-OL1   | 0    | 5.4598    | 0.7402 | 0.6229    | 0.6331 | 0.6247      | 0.9842 |
| c17 | A4-OL1   | 0.01 | 5.4602    | 0.7401 | 0.6239    | 0.6191 | 0.6108      | 1.0054 |
| c18 | A4-OL1   | 0.05 | 5.4922    | 0.7401 | 0.6236    | 0.5706 | 0.5628      | 1.0885 |
| c19 | A4-OL1   | 0.1  | 5.5504    | 0.7399 | 0.6233    | 0.5325 | 0.5249      | 1.1674 |
| c20 | A4-OL1   | 0.5  | 6.0403    | 0.7394 | 0.6227    | 0.4591 | 0.4511      | 1.3542 |
| c21 | A7       | 0    | 5.4704    | 0.7400 | 0.6227    | 0.6339 | 0.6260      | 0.9824 |
| c22 | A7       | 0.01 | 5.4611    | 0.8460 | 0.6212    | 0.6382 | 0.6306      | 0.9734 |
| c23 | A7       | 0.05 | 5.4405    | 0.8458 | 0.6208    | 0.6352 | 0.6267      | 0.9773 |
| c24 | A7       | 0.1  | 5.4309    | 0.8453 | 0.6209    | 0.5934 | 0.5858      | 1.0443 |
| c25 | A7       | 0.5  | 5.7076    | 0.8449 | 0.6211    | 0.4789 | 0.4714      | 1.2955 |
| c26 | A7-OL1   | 0    | 5.4824    | 0.7400 | 0.6231    | 0.6337 | 0.6255      | 0.9831 |
| c27 | A7-OL1   | 0.01 | 5.4778    | 0.8456 | 0.6213    | 0.6384 | 0.6308      | 0.9725 |
| c28 | A7-OL1   | 0.05 | 5.4651    | 0.8459 | 0.6214    | 0.6346 | 0.6274      | 0.9782 |
| c29 | A7-OL1   | 0.1  | 5.4315    | 0.8459 | 0.6210    | 0.5920 | 0.5847      | 1.0473 |
| c30 | A7-OL1   | 0.5  | 5.8313    | 0.8441 | 0.6210    | 0.4731 | 0.4679      | 1.3112 |

Table 14: Descriptive association sensitivity. Each column regresses its timing ratio on canonical FP16 model-wide sparsity with an intercept. Native relative uses native/K050 latency; sparse factor uses the incremental definition above. Subsets are retrospective and overlapping, with no independent training seeds.

| Checkpoint subset         | $n$ | Native-relative $R^2$ | Sparse-factor $R^2$ |
|---------------------------|-----|-----------------------|---------------------|
| All checkpoints           | 30  | 0.8167                | 0.5037              |
| Within A1-H-L1            | 4   | 0.8944                | 0.8908              |
| Within A1-H-OL1           | 4   | 0.8888                | 0.8921              |
| Within A4                 | 5   | 0.8353                | 0.8346              |
| Within A4-OL1             | 5   | 0.9087                | 0.9095              |
| Within A7                 | 5   | 0.9528                | 0.9356              |
| Within A7-OL1             | 5   | 0.9403                | 0.9825              |
| Without A0                | 29  | 0.8104                | 0.4943              |
| Without A1-H              | 29  | 0.8101                | 0.4989              |
| Without A1-H-L1           | 26  | 0.7937                | 0.5077              |
| Without A1-H-OL1          | 26  | 0.7929                | 0.5007              |
| Without A4                | 25  | 0.8497                | 0.5685              |
| Without A4-OL1            | 25  | 0.8319                | 0.5322              |
| Without A7                | 25  | 0.8401                | 0.4977              |
| Without A7-OL1            | 25  | 0.8520                | 0.4910              |
| Without search checkpoint | 29  | 0.8521                | 0.4080              |

## D.8 EVIDENCE AND DATA ACCOMPANYING THE DRAFT

The supplementary directory `supplementary-data/` contains all 54 training records with integer operation counts, all 540 clipping evaluations and frontier memberships, seven signed histograms with zero and tail counts, figure coordinates, and the 30-checkpoint runtime reduction. The included README defines fields and units. Copy hashes are recorded in `SOURCES.json`; `protocol.json` gives data preparation, training settings and implementation identities. Paths refer to the accompanying repository; the supplementary directory is not a standalone checkpoint or kernel release. The `revision-audit/` subdirectory adds the retrospective quality-budget reduction, absolute runtime summaries and sensitivity checks, per-layer projection-input row statistics, and historical-pressure identity audit. Raw identifiers remain unchanged:  $\mathcal{S}_{\text{model}}$  corresponds to `R_model`,  $\mathcal{S}_{\text{block}}$  to `R_block`, and architectural reach  $R_{\text{arch}}$  to the existing `R_model_max` fields.

The accompanying repository includes the figure and table scripts. The Analysis 019 and 020 READMEs give commands for regenerating the revised visuals from retained numerical records, separate from commands that require original raw run artifacts. Analysis 020 adds the OL1 cap audit, matched ReLU/clipping comparisons and raw training trajectories; `revision-v2/` contains their claim ledger, derived data and verification record. The draft README gives the LaTeX build commands. Checkpoint weights and runtime compilation dependencies are not bundled in the supplementary measurement directory; these commands do not claim to reproduce training or GPU measurements from that directory alone.